

CM515 Day 2: Plotting with ggplot()

1. Putting it all together

We are going to take some of my data and combine the principles from day1 of the ggplot section to make an awesome graph!

Context:

I have data for a bunch of strains where I measured the growth (as optical density) every hour for 26 hours in quadruplicate.

This experiment was done at 85C and 95C

The strains are:

TS559-Parent Strain TK0887-deleted for Endonuclease Q TK1898-deleted for Endonuclease MS TK2143-deleted for Uracil DNA Glycosylase TK0887/TK1898 and TK1898/TK2143-deleted for both genes TK0887/TK1898/TK2143-deleted for all three genes

```
####  
#01#  
####
```

```
dU_GC<-read.csv("Growth_Curve_dU.csv",fill=TRUE,header=TRUE)
```

```
head(dU_GC)
```

```
##   Hours Blank.1 Blank.2 Blank.3 Blank.4 TS559.1 TS559.2 TS559.3 TS559.4  
## 1     0   0.001  -0.035  -0.047   0.000  -0.003   0.018  -0.017   0.000  
## 2     1  -0.023  -0.030  -0.036  -0.051   0.049   0.020   0.023   0.032  
## 3     2  -0.043  -0.033  -0.041  -0.035   0.093   0.082   0.074   0.055  
## 4     3  -0.045  -0.054  -0.045  -0.034   0.144   0.148   0.133   0.129  
## 5     4  -0.045  -0.044  -0.045  -0.035   0.269   0.245   0.252   0.243  
## 6     5  -0.042  -0.050  -0.046  -0.036   0.393   0.389   0.409   0.397  
##   TK0887.1 TK0887.2 TK0887.3 TK0887.4 TK1898.1 TK1898.2 TK1898.3 TK1898.4  
## 1  -0.011  -0.011  -0.008  -0.018   0.000   0.004  -0.015  -0.013  
## 2   0.050   0.034   0.021   0.033   0.061   0.037   0.017   0.020  
## 3   0.090   0.086   0.096   0.076   0.056   0.061   0.073   0.091  
## 4   0.162   0.158   0.153   0.146   0.168   0.111   0.130   0.145  
## 5   0.253   0.256   0.276   0.259   0.239   0.199   0.237   0.247  
## 6   0.360   0.350   0.382   0.371   0.316   0.314   0.386   0.404  
##   TK2143.1 TK2143.2 TK2143.3 TK2143.4 TK0887.TK1898.1 TK0887.TK1898.2  
## 1  -0.009  -0.015   0.040   0.004                -0.012                -0.007  
## 2   0.007   0.031   0.022   0.037                0.027                0.017  
## 3   0.066   0.072   0.069   0.062                0.068                0.080
```

## 4	0.094	0.125	0.121	0.125	0.130	0.140
## 5	0.179	0.246	0.231	0.226	0.310	0.225
## 6	0.314	0.398	0.399	0.404	0.310	0.324
##	TK0887.TK1898.3	TK0887.TK1898.4	TK0887.TK2143.1	TK0887.TK2143.2		
## 1		0.002	0.012	0.003		-0.025
## 2		0.022	0.043	-0.001		0.006
## 3		0.077	0.082	0.037		0.037
## 4		0.073	0.080	0.070		0.107
## 5		0.245	0.294	0.133		0.143
## 6		0.344	0.468	0.232		0.267
##	TK0887.TK2143.3	TK0887.TK2143.4	TK1898.TK2143.1	TK1898.TK2143.2		
## 1		-0.017	-0.009	-0.019		-0.009
## 2		0.006	0.003	0.000		0.006
## 3		0.033	0.035	0.029		0.032
## 4		0.073	0.080	0.063		0.075
## 5		0.144	0.153	0.113		0.124
## 6		0.256	0.274	0.219		0.246
##	TK1898.TK2143.3	TK1898.TK2143.4	TK0887.TK1898.TK2143.1	TK0887.TK1898.TK2143.2		
## 1		-0.020	0.047	0.008		-0.025
## 2		0.010	0.073	0.030		-0.003
## 3		0.039	0.043	0.050		0.021
## 4		0.079	0.134	0.072		0.053
## 5		0.137	0.142	0.073		0.088
## 6		0.260	0.308	0.092		0.092
##	TK0887.TK1898.TK2143.3	TK0887.TK1898.TK2143.4				
## 1		-0.001		-0.016		
## 2		0.007		0.001		
## 3		0.035		0.030		
## 4		0.079		0.064		
## 5		0.068		0.090		
## 6		0.086		0.106		

2. What happens when you use not tidy data.

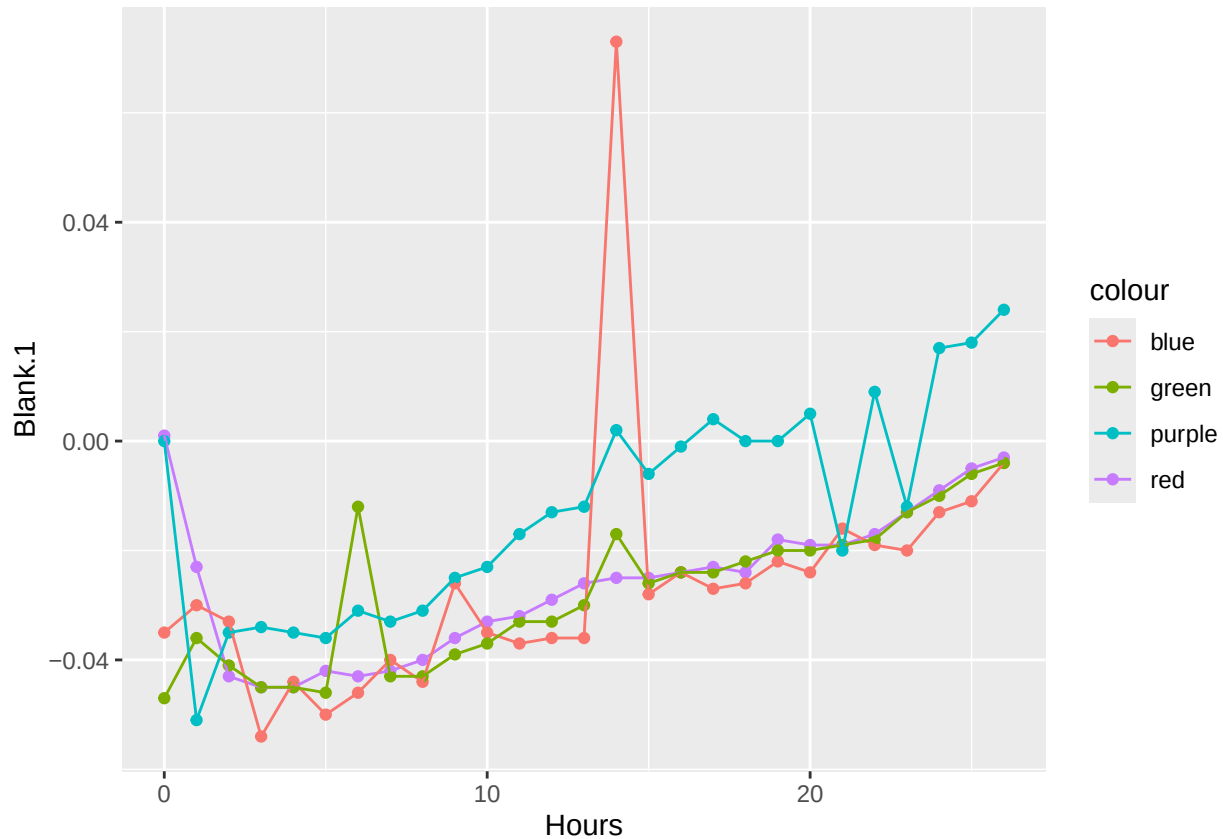
This data is not tidy, which means that we will have a lot lot lot of issues.

1. Because the strains and replicates are separated, I can only graph one strain at a time, one replicate at a time.

```
####
#02#
####

dU_GC%>%
  ggplot() +
  geom_point(aes(x=Hours, y=Blank.1, color="red")) +
  geom_point(aes(x=Hours, y=Blank.2, color="blue")) +
  geom_point(aes(x=Hours, y=Blank.3, color="green")) +
  geom_point(aes(x=Hours, y=Blank.4, color="purple")) +
  geom_line(aes(x=Hours, y=Blank.1, color="red")) +
```

```
geom_line(aes(x=Hours, y=Blank.2, color="blue")) +  
geom_line(aes(x=Hours, y=Blank.3, color="green")) +  
geom_line(aes(x=Hours, y=Blank.4, color="purple"))
```



2. What happens when you do statistics on not tidy data

I want to graph the average ODs for each strain at every hour.

If I am using the current data in not tidy format, this is the amount of statistical analysis I have to do:

1. Pivot the data longer and make individual tables for each strain that summarize the data. I basically have to make tidy data using R which, while doable, is a lot of unnecessary work.
2. I then have to combine every single table for each individual strain together using rbind.

```
####  
#02#  
####  
  
GC_85_Blank <-dU_GC %>%  
pivot_longer(cols = starts_with("Blank"),  
             names_to = "Blank Replicates",  
             values_to = "Blank Absorbance",
```

```

        values_drop_na = TRUE,
        values_transform = list(Absorbance=as.character)) %>%
select(c(Hours, `Blank Replicates`, `Blank Absorbance`)) %>%
group_by(Hours) %>%
summarize(mean = mean(`Blank Absorbance`), sd= sd(`Blank Absorbance`)) %>%
add_column(Strain = "Blank")

GC_85_TS559 <- dU_GC %>%
  pivot_longer(cols = starts_with("TS559"),
    names_to = "TS559 Replicates",
    values_to = "TS559 Absorbance",
    values_drop_na = TRUE) %>%
  select(c(Hours, `TS559 Replicates`, `TS559 Absorbance`)) %>%
  group_by(Hours) %>%
  summarize(mean = mean(`TS559 Absorbance`), sd = sd(`TS559 Absorbance`)) %>%
  add_column(Strain = "TS559")

GC_85_0887 <- dU_GC %>%
  pivot_longer(cols = starts_with("TK0887"),
    names_to = "0887 Replicates",
    values_to = "0887 Absorbance",
    values_drop_na = TRUE) %>%
  select(c(Hours, `0887 Replicates`, `0887 Absorbance`)) %>%
  group_by(Hours) %>%
  summarize(mean = mean(`0887 Absorbance`), sd = sd(`0887 Absorbance`)) %>%
  add_column(Strain = "TK0887-Endonuclease Q")

GC_85_1898 <- dU_GC %>%
  pivot_longer(cols = starts_with("TK1898"),
    names_to = "1898 Replicates",
    values_to = "1898 Absorbance",
    values_drop_na = TRUE) %>%
  select(c(Hours, `1898 Replicates`, `1898 Absorbance`)) %>%
  group_by(Hours) %>%
  summarize(mean = mean(`1898 Absorbance`), sd = sd(`1898 Absorbance`)) %>%
  add_column(Strain = "TK1898-Endonuclease MS")

GC_85_2143 <- dU_GC %>%
  pivot_longer(cols = starts_with("TK2143"),
    names_to = "2143 Replicates",
    values_to = "2143 Absorbance",
    values_drop_na = TRUE) %>%
  select(c(Hours, `2143 Replicates`, `2143 Absorbance`)) %>%
  group_by(Hours) %>%
  summarize(mean = mean(`2143 Absorbance`), sd = sd(`2143 Absorbance`)) %>%
  add_column(Strain = "TK2143-Uracil DNA Glycosylase")

GC_85_0887_1898 <- dU_GC %>%
  pivot_longer(cols = starts_with("TK0887.TK1898"),
    names_to = "TK0887/TK1898 Replicates",
    values_to = "TK0887/TK1898 Absorbance",
    values_drop_na = TRUE) %>%

```

```

select(c(Hours, `TK0887/TK1898 Replicates`, `TK0887/TK1898 Absorbance`)) %>%
group_by(Hours) %>%
summarize(mean = mean(`TK0887/TK1898 Absorbance`),
          sd = sd(`TK0887/TK1898 Absorbance`))%>%
add_column(Strain = "TK0887/TK1898")

GC_85_0887_2143 <- dU_GC %>%
  pivot_longer(cols = starts_with("TK0887.TK2143"),
               names_to = "TK0887/TK2143 Replicates",
               values_to = "TK0887/TK2143 Absorbance",
               values_drop_na = TRUE) %>%
  select(c(Hours, `TK0887/TK2143 Replicates`, `TK0887/TK2143 Absorbance`)) %>%
  group_by(Hours) %>%
  summarize(mean = mean(`TK0887/TK2143 Absorbance`),
            sd = sd(`TK0887/TK2143 Absorbance`))%>%
  add_column(Strain = "TK0887/TK2143")

GC_85_1898_2143 <- dU_GC %>%
  pivot_longer(cols = starts_with("TK1898.TK2143"),
               names_to = "TK1898/TK2143 Replicates",
               values_to = "TK1898/TK2143 Absorbance",
               values_drop_na = TRUE) %>%
  select(c(Hours, `TK1898/TK2143 Replicates`, `TK1898/TK2143 Absorbance`)) %>%
  group_by(Hours) %>%
  summarize(mean = mean(`TK1898/TK2143 Absorbance`),
            sd = sd(`TK1898/TK2143 Absorbance`))%>%
  add_column(Strain = "TK1898/TK2143")

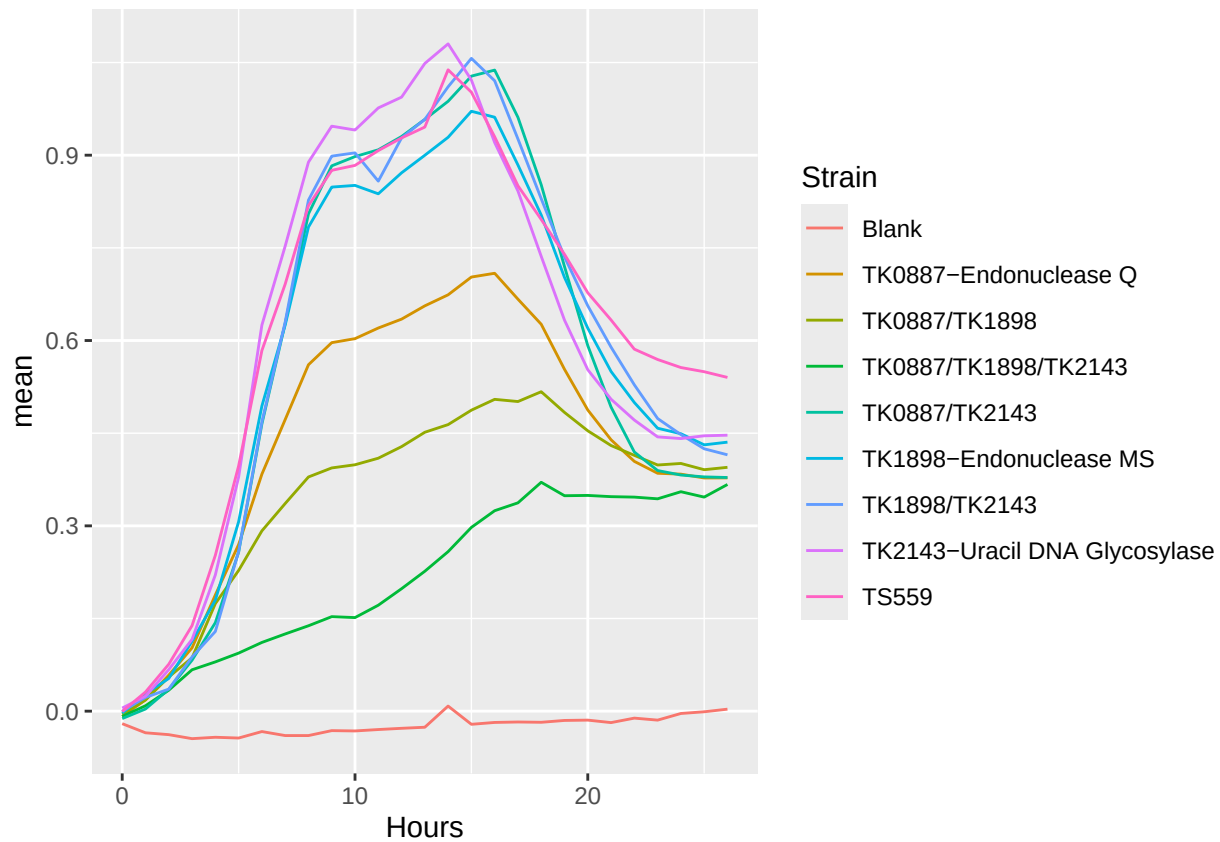
GC_85_0887_1898_2143<- dU_GC %>%
  pivot_longer(cols = starts_with("TK0887.TK1898.TK2143"),
               names_to = "TK0887/TK1898/TK2143 Replicates",
               values_to = "TK0887/TK1898/TK2143 Absorbance",
               values_drop_na = TRUE) %>%
  select(c(Hours, `TK0887/TK1898/TK2143 Replicates`,
               `TK0887/TK1898/TK2143 Absorbance`)) %>%
  group_by(Hours) %>%
  summarize(mean = mean(`TK0887/TK1898/TK2143 Absorbance`),
            sd = sd(`TK0887/TK1898/TK2143 Absorbance`))%>%
  add_column(Strain = "TK0887/TK1898/TK2143")

# combine tables
dU_Table <-rbind(GC_85_Blank, GC_85_TS559)
dU_Table <-rbind(dU_Table, GC_85_0887)
dU_Table <-rbind(dU_Table, GC_85_1898)
dU_Table <-rbind(dU_Table, GC_85_2143)
dU_Table <-rbind(dU_Table, GC_85_0887_1898)
dU_Table <-rbind(dU_Table, GC_85_0887_2143)
dU_Table <-rbind(dU_Table, GC_85_1898_2143)
dU_Table_Final <-rbind(dU_Table, GC_85_0887_1898_2143)

dU_Table_Final %>%

```

```
ggplot(aes(x=Hours, y=mean, color=Strain)) +  
  geom_line()
```



Not only is that an insane amount of work (took me several days to figure out) and a lot of lines of code, the graph is factually incorrect.

The lines of each strain are not exactly correct.

Because TK0887 is a repeating column, the 'startswith' argument cannot differentiate between each column that has the name TK0887.

This means way way more work, more time being frustrated, and less time making your graph awesome!

Let's make the data tidy, and then start over.

3. Working with tidy data. Mapping Aesthetics

We can complete the same statistical analysis in a SIGNIFICANTLY shorter time span.

Try plotting mean (optical density) against time now, and factor the color by strain.

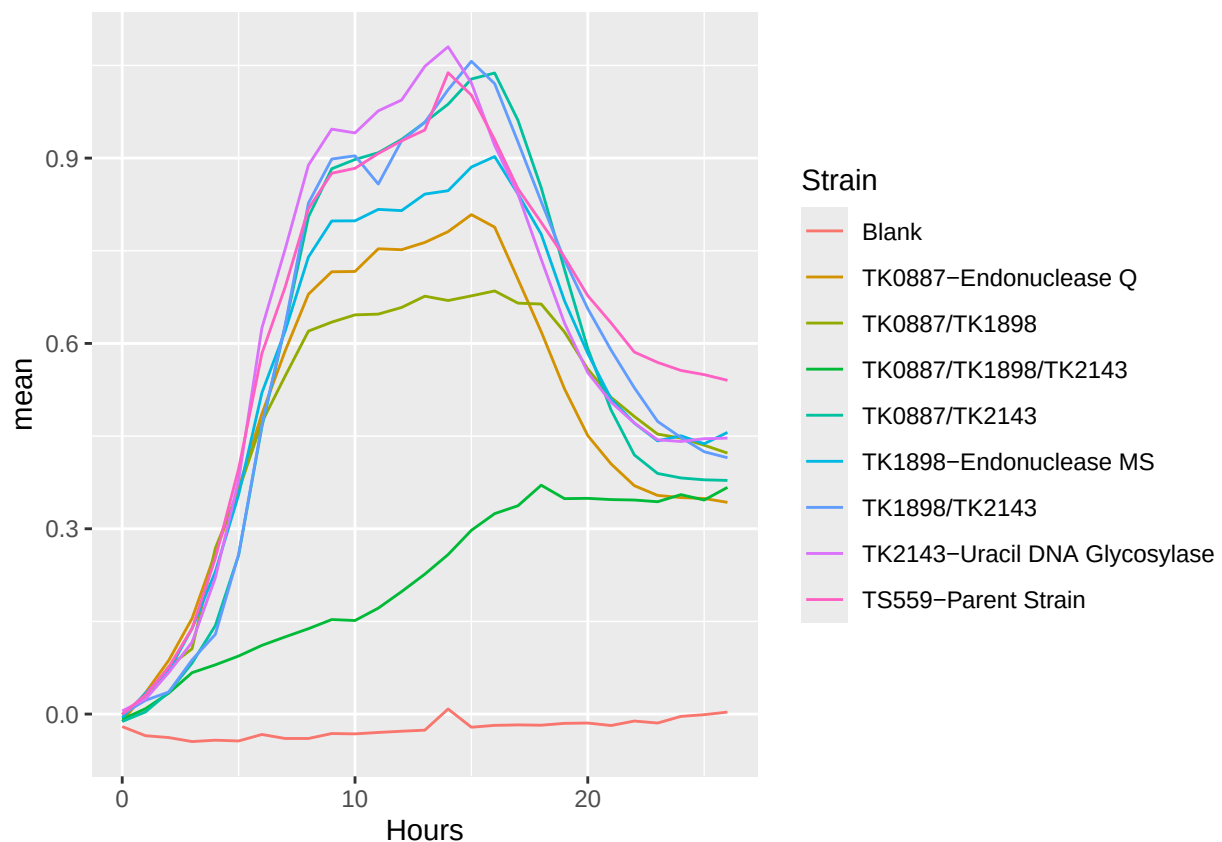
```
####  
#03#  
####  
  
dU_GC_tidy<-read.csv("Growth_Curve_dU_tidy.csv",fill=TRUE,header=TRUE)
```

```
head(dU_GC_tidy)
```

```
##   Hours Strain Replicates      OD  X  X.1 X.2 X.3 X.4 X.5 X.6 X.7 X.8
## 1     0 Blank           1  0.001 NA   NA   NA   NA   NA   NA   NA   NA
## 2     1 Blank           1 -0.023 NA   NA   NA   NA   NA   NA   NA   NA
## 3     2 Blank           1 -0.043 NA   NA   NA   NA   NA   NA   NA   NA
## 4     3 Blank           1 -0.045 NA   NA   NA   NA   NA   NA   NA   NA
## 5     4 Blank           1 -0.045 NA   NA   NA   NA   NA   NA   NA   NA
## 6     5 Blank           1 -0.042 NA   NA   NA   NA   NA   NA   NA   NA
```

```
dU_GC_tidy_Stat <- dU_GC_tidy %>%
  group_by(Hours, Strain) %>%
  summarize(mean = mean(OD), sd = sd(OD), n=n(), se = sd/sqrt(n)) %>%
  group_by(Strain)

dU_GC_tidy_Stat %>%
  ggplot(aes(x = Hours, y = mean, color=Strain)) +
  geom_line()
```



4. Factoring the Mapping

Our legend labels do not say in any way that these are gene deletions. I did not include deltas because R would not interpret this from a csv file.

I recommend always fixing the names of columns, Rows, and categorical variables before graphing because afterwards it is a lot of code to fix those issues.

We should put a delta sign in front of each gene number, but how?

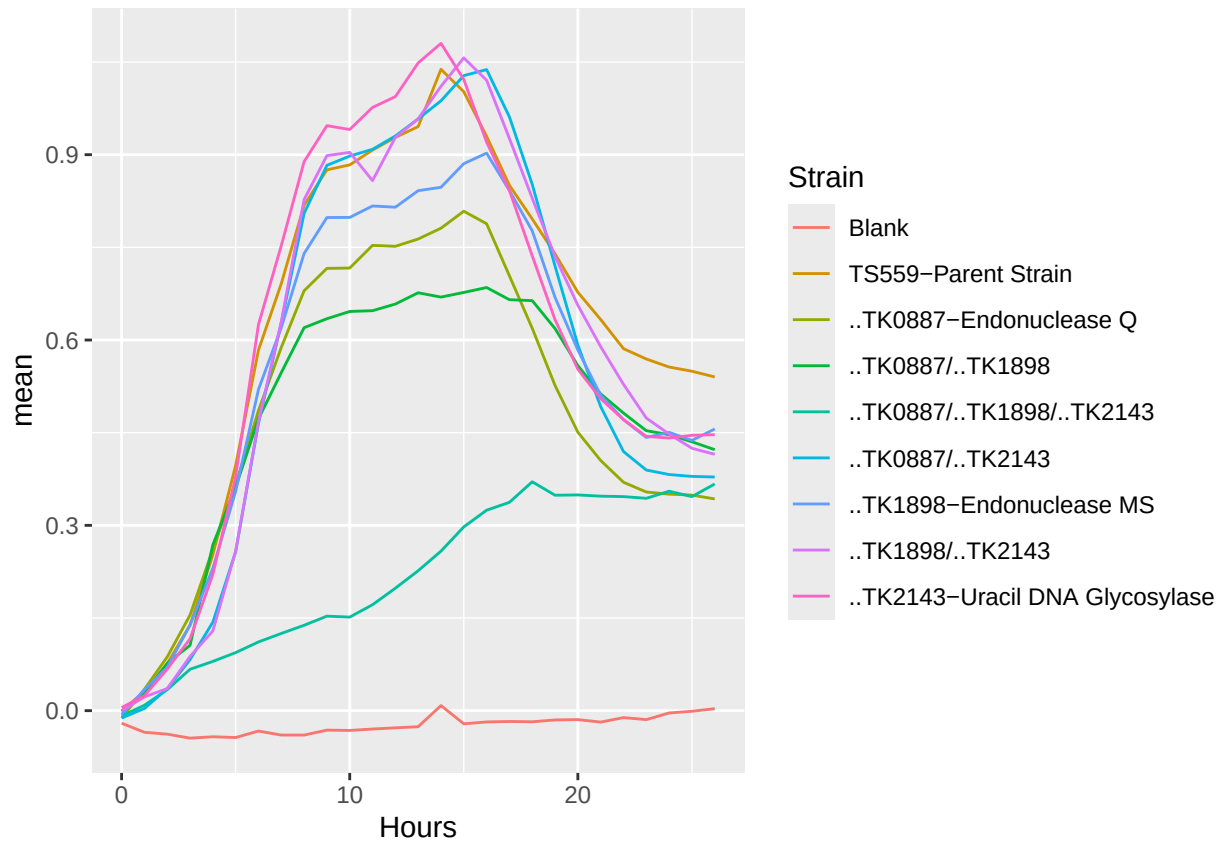
Fortunately, R can interpret unicode.

Using labels, we can rename our legend to include deltas in front of each gene.

https://en.wikipedia.org/wiki/List_of_Unicode_characters

You can find all greek letters on wikipedia using the link above under Greek and Coptic, or just ask ChatGPT. We put the unicode right into the string (unicode for delta is `0394`).

```
####  
#04#  
####  
  
dU_GC_tidy_Stat <- dU_GC_tidy_Stat %>%  
  mutate(Strain =  
    recode(Strain, "TK0887-Endonuclease Q" = "\u0394TK0887-Endonuclease Q",  
               "TK1898-Endonuclease MS" = "\u0394TK1898-Endonuclease MS",  
               "TK2143-Uracil DNA Glycosylase" = "\u0394TK2143-Uracil DNA Glycosylase",  
               "TK0887/TK1898" = "\u0394TK0887/\u0394TK1898",  
               "TK0887/TK2143" = "\u0394TK0887/\u0394TK2143",  
               "TK1898/TK2143" = "\u0394TK1898/\u0394TK2143",  
               "TK0887/TK1898/TK2143" = "\u0394TK0887/\u0394TK1898/\u0394TK2143"))  
  
dU_GC_tidy_Stat%>%  
  ggplot(aes(x=Hours, y=mean, color=Strain)) +  
  geom_line()
```

5. Picking the right geometry

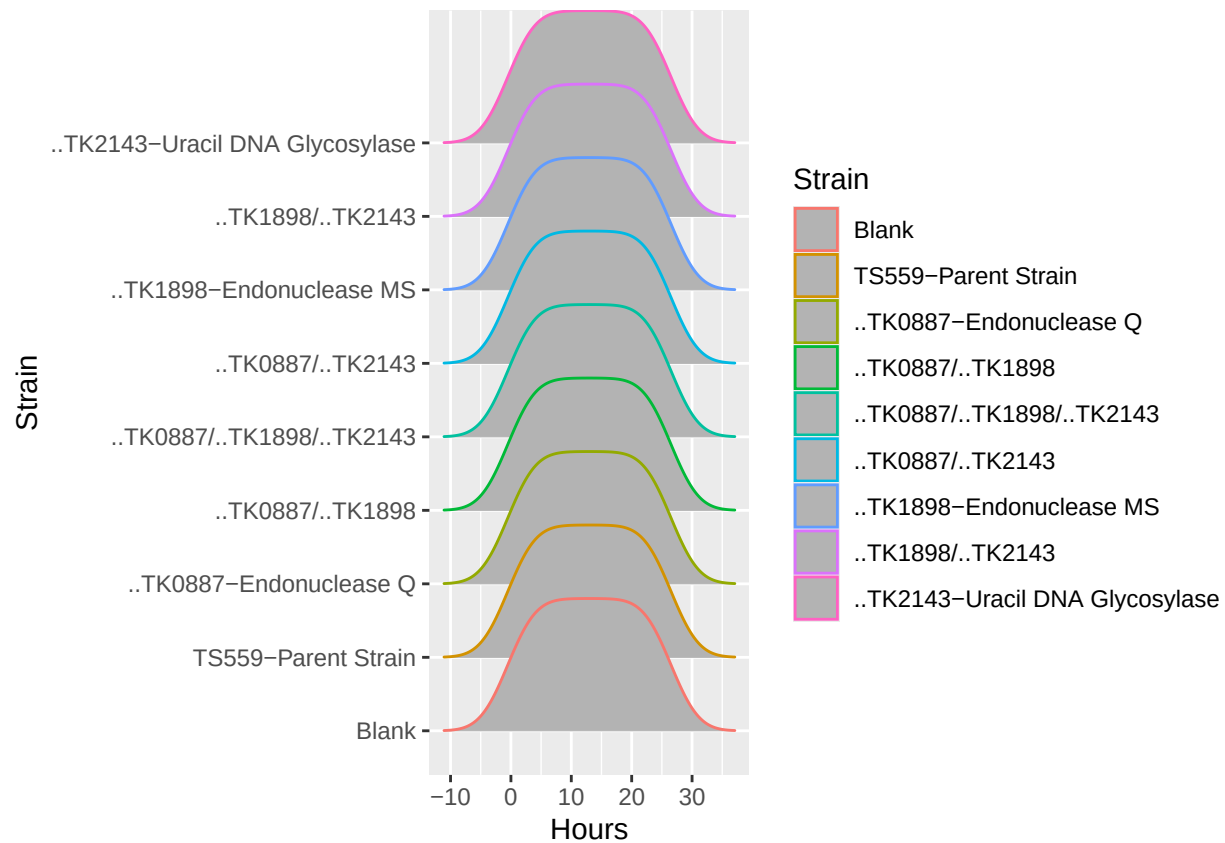
Now the data has received proper statistics and been mapped.

We can now tell the dots apart, but, maybe a scatter plot is not the best way to represent this data.

Lets look at a few different geometries.

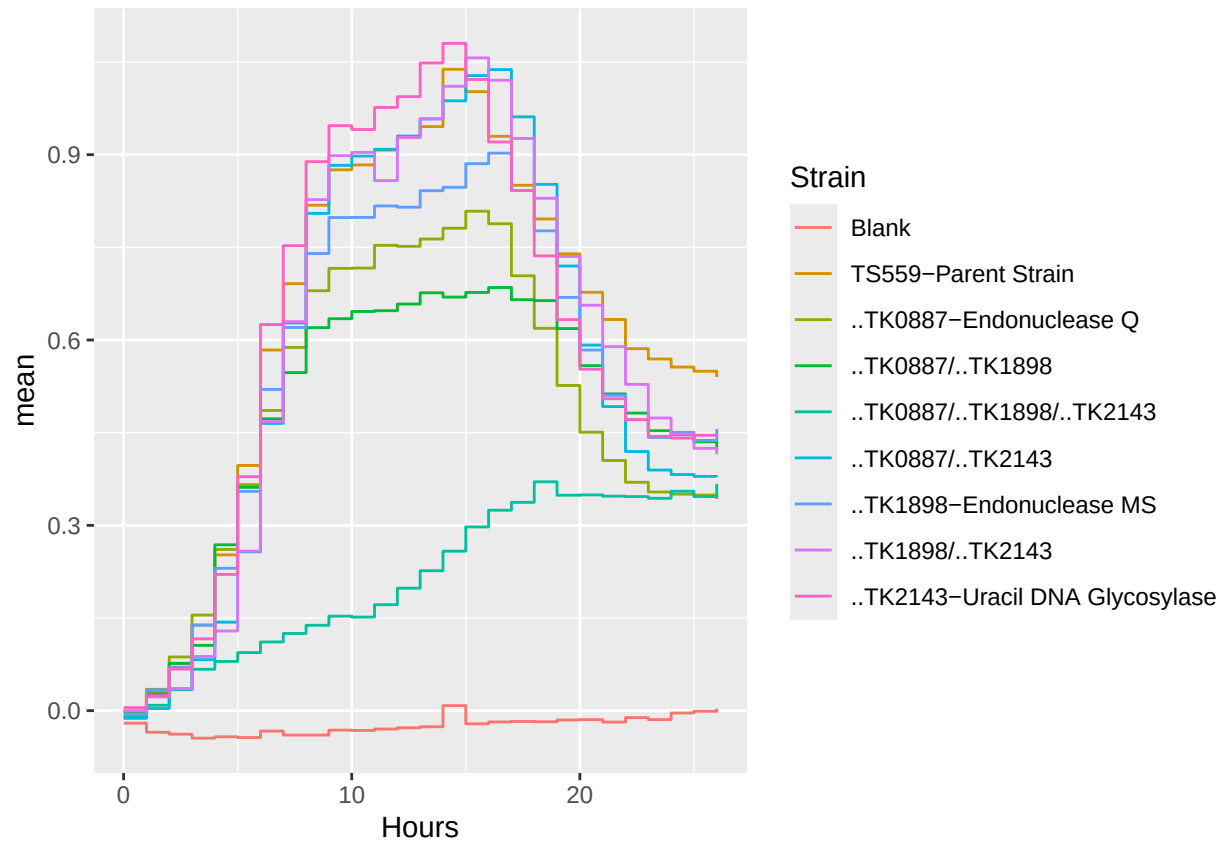
```
####
#06#
####

dU_GC_tidy_Stat%>%
  ggplot(aes(x=Hours, y= Strain ,color = Strain)) +
  geom_density_ridges_gradient()
```

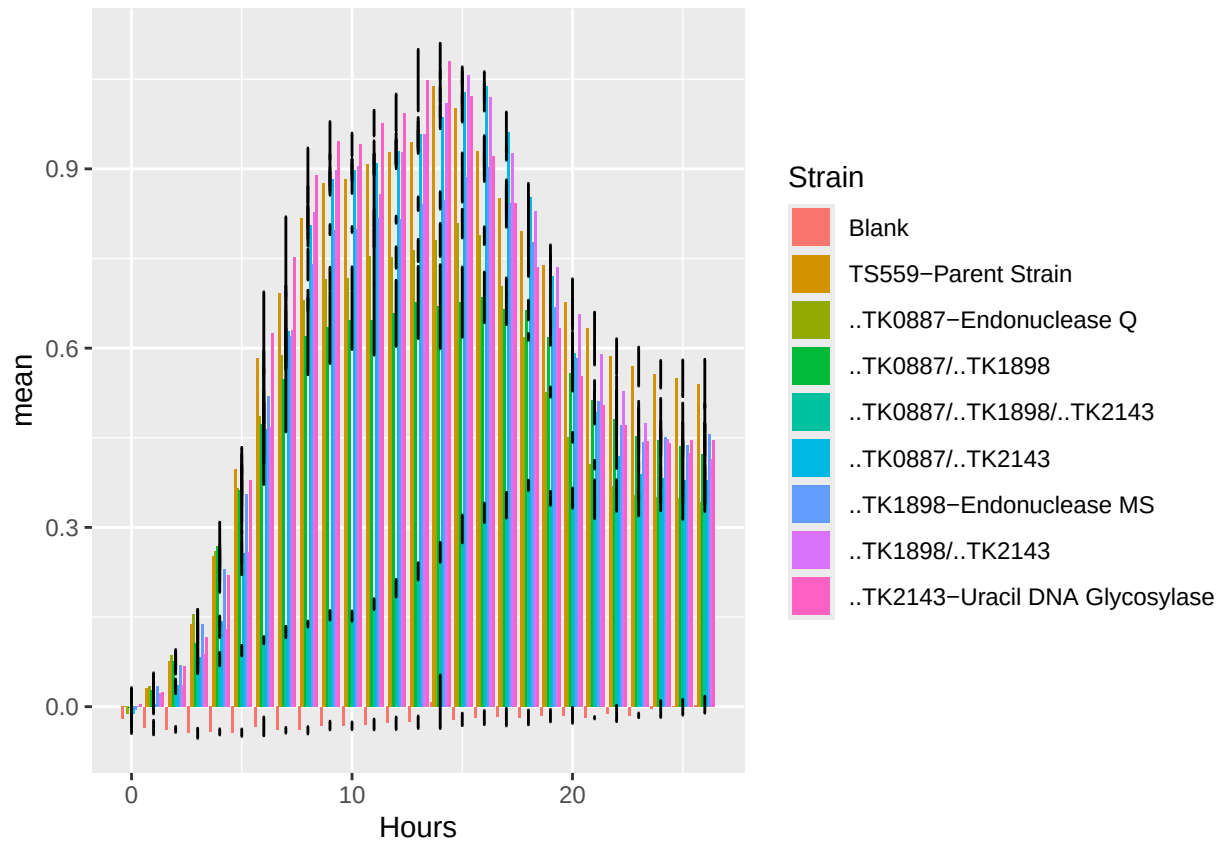


The {ggridges} package draws density curves which are by definition scaled to an area of 1. The height of the ridges therefore cannot be mapped to the mean OD so we must use a different graph.

```
dU_GC_tidy_Stat%>%
  ggplot(aes(x=Hours, y=mean, color=Strain, fill=Strain)) +
  geom_step()
```

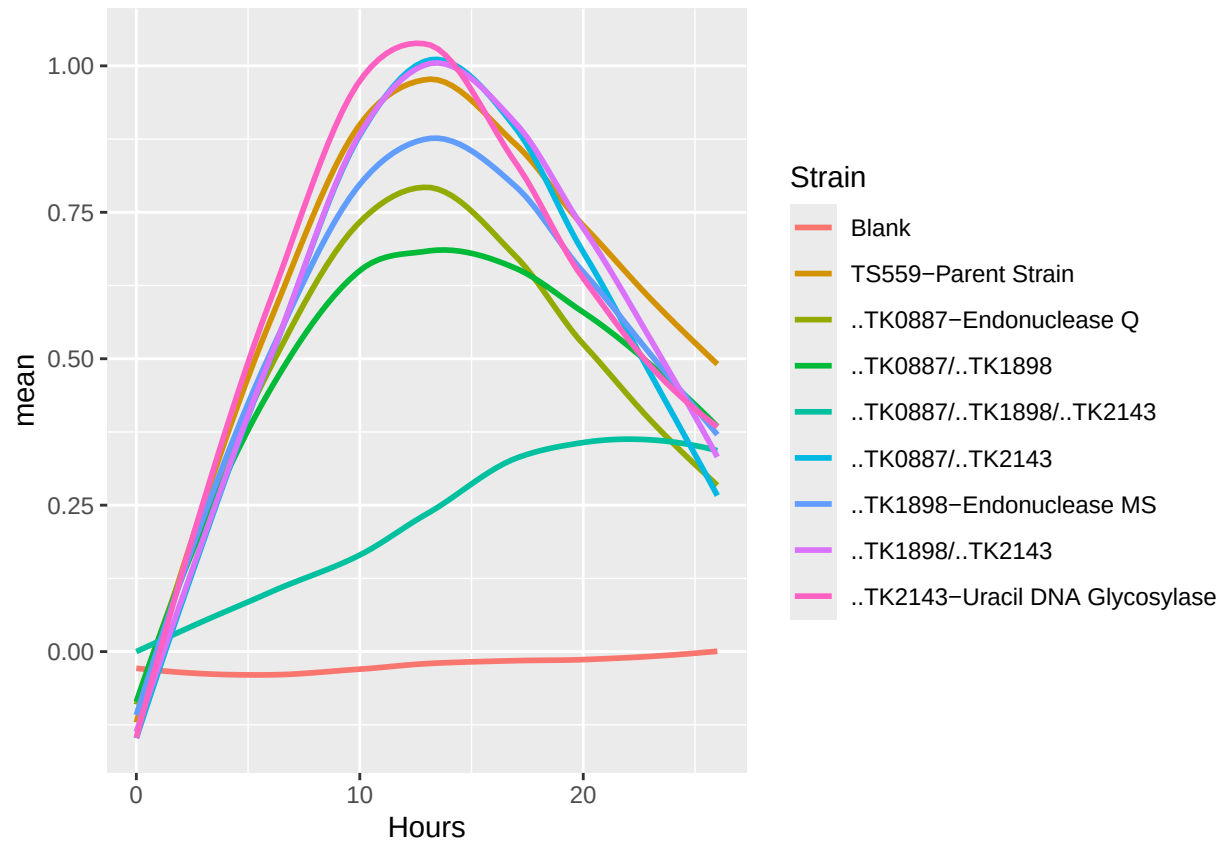


```
dU_GC_tidy_Stat%>%
  ggplot(aes(x=Hours, y=mean, fill=Strain, ymax=mean+sd, ymin=mean-sd)) +
  geom_col(size=1.2, position=position_dodge(preserve='single')) +
  geom_errorbar(width=0.1)
```



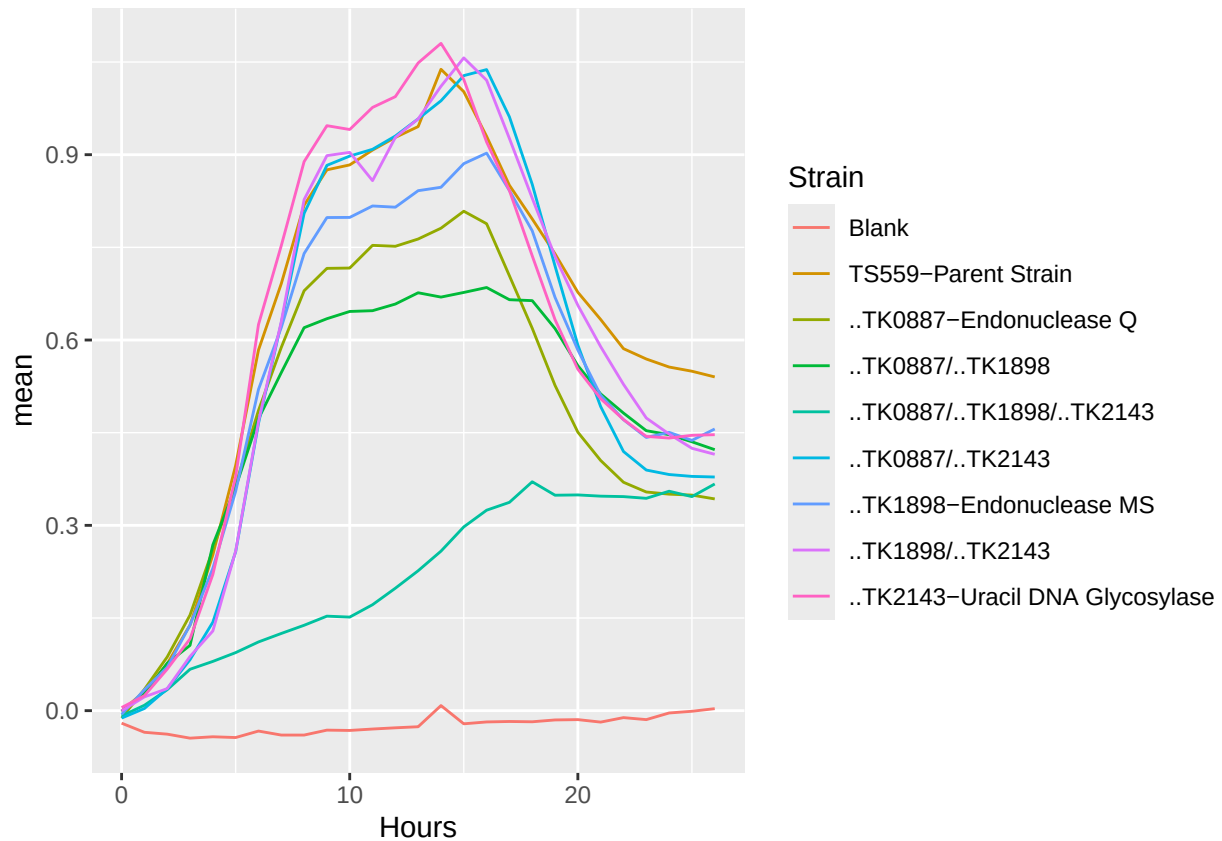
Makes a step-wise line go through each point, but in this context it is quite ugly. `Geom_col` is... just... no...

```
dU_GC_tidy_Stat%>%
  ggplot(aes(x=Hours, y=mean, color=Strain, fill=Strain)) +
  geom_smooth(se=FALSE)
```



This one is pretty great when you are only measuring growth for a few strains, but with nine it looks overwhelming.

```
dU_GC_tidy_Stat%>%
  ggplot(aes(x=Hours, y=mean, color=Strain)) +
  geom_line()
```



#Perfect!

5b. Overlapping Geometries

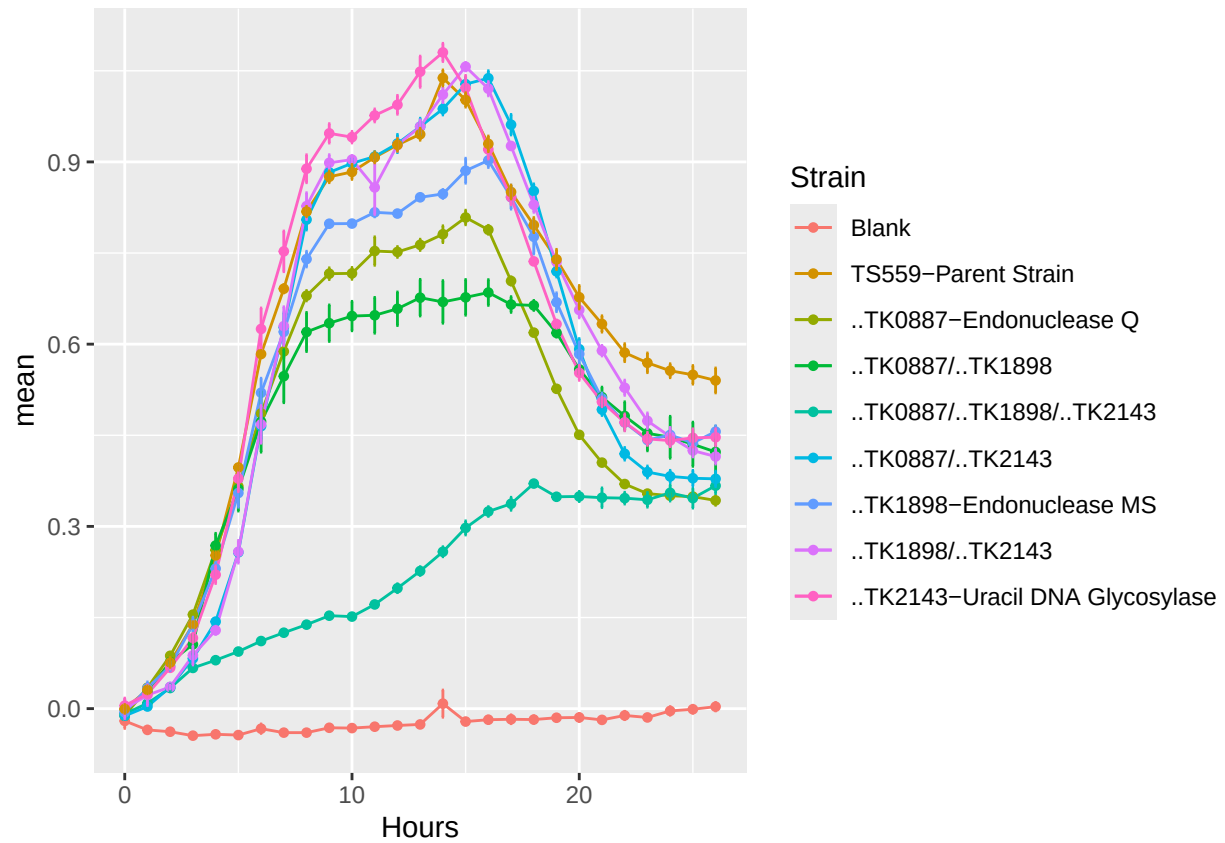
While the line follows the trend, it is not a bad idea to include the points where the data are.

We can also include error bars. Keep in mind that standard deviation is the deviation of your sample values from your mean while standard error measures the uncertainty of your estimated mean (used to establish confidence intervals). For error bars we often want to use `se`.

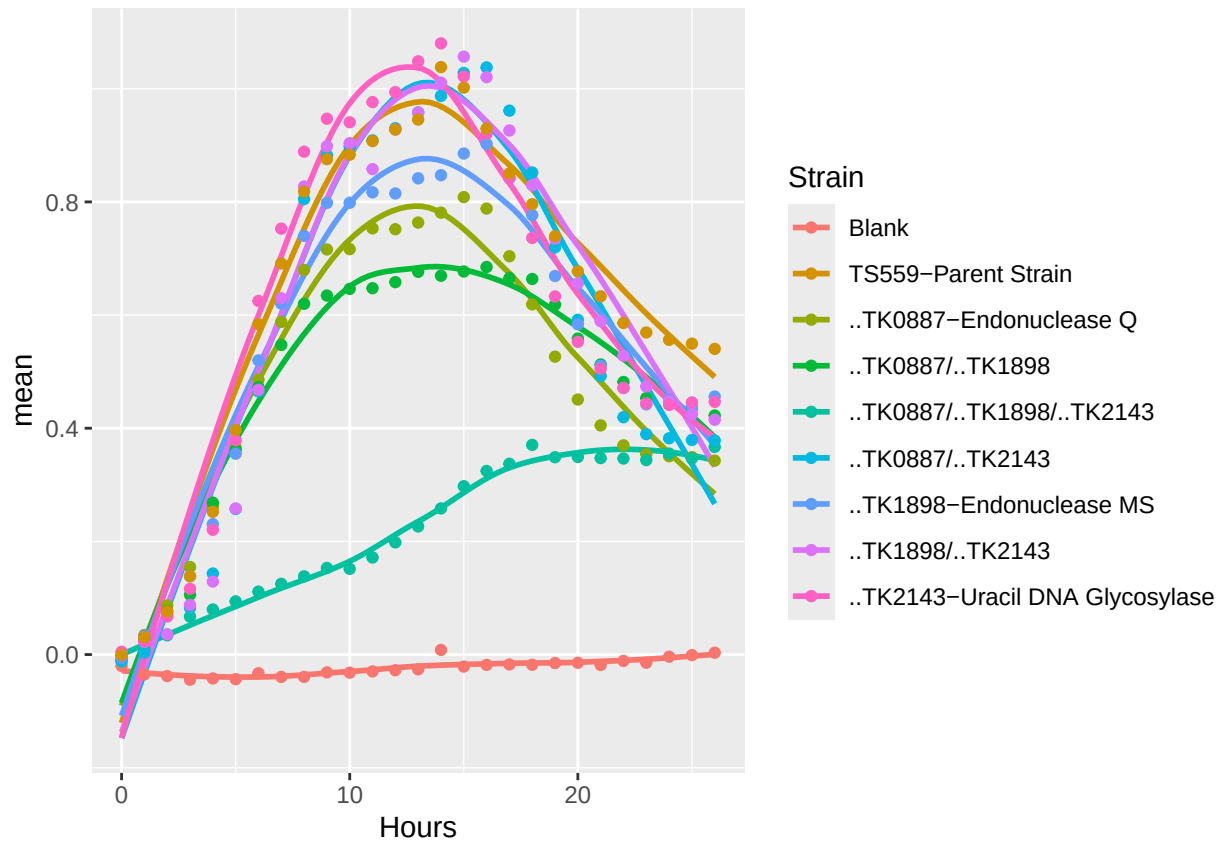
Hypothetically, we could use `geom_smooth` for this, and just remove the standard error line. Unfortunately, the smooth function likes to smooth the line, so the points do not exactly match up.

```
####
#05b#
####

dU_GC_tidy_Stat%>%
  ggplot(aes(x=Hours, y=mean, color=Strain, ymax=mean+se, ymin=mean-se)) +
  geom_line(size=0.5) +
  geom_point(size=1.2) +
  geom_errorbar(width=0.1)
```



```
dU_GC_tidy_Stat%>%
  ggplot(aes(x=Hours, y=mean, color=Strain)) +
  geom_smooth(se=FALSE) +
  geom_point()
```



6. Labels

Now this graph is starting to display our data appropriately.

Let's change the axis labels

We want to specify that the y axis is optical density at 600nm.

X axis is Time in hours

Bquote allows us to customize our text to be bold, italicized and underlined. We can also add superscripts, subscripts, and many other components.

For more on bquote-use `help("plotmath")` or ask ChatGPT.

```
####
#06#
####

#help("plotmath")

dU_GC_tidy_Stat%>%
  ggplot(aes(x=Hours, y=mean, color=Strain, ymax=mean+se, ymin=mean-se)) +
  geom_line(size=0.5) +
  geom_point(size=1.2) +
  geom_errorbar(width=0.1) +
  labs(title = "Strains lacking all dU repair components have the most severe fitness defect", subtitle=
```

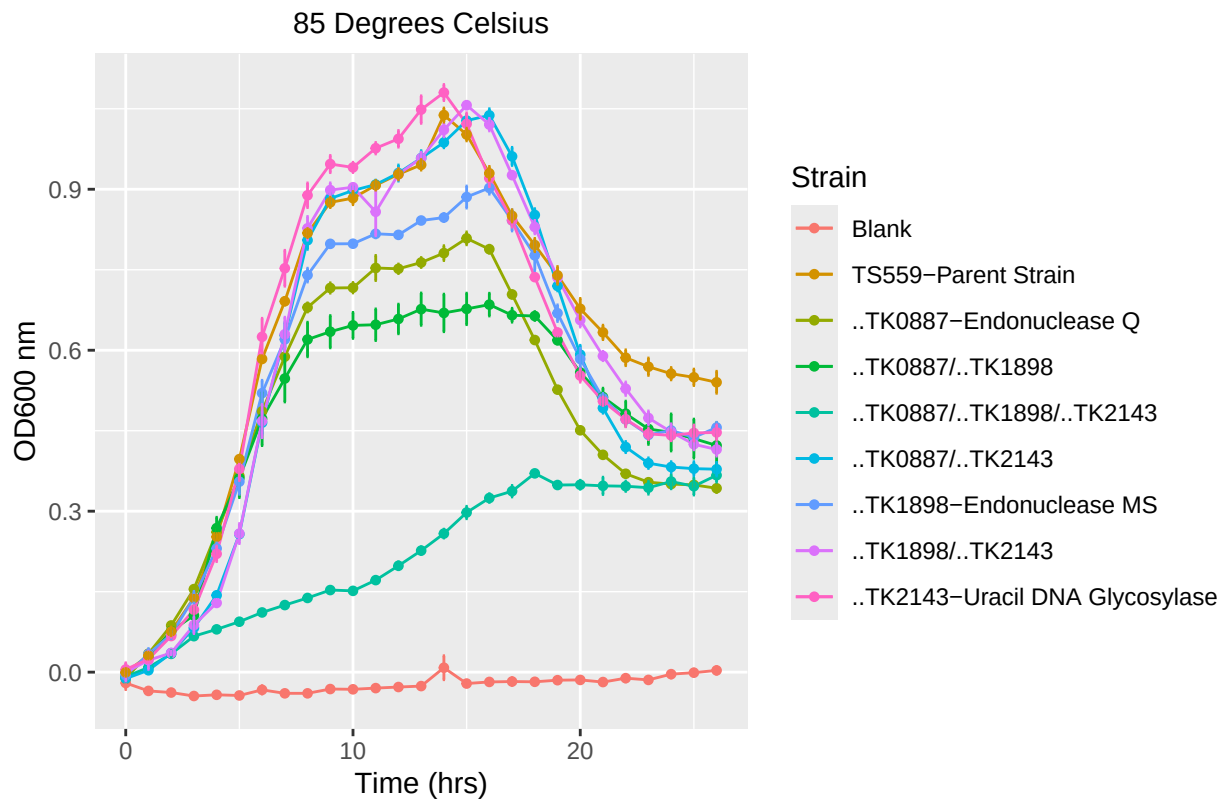


```

y = "OD600 nm",
x = "Time (hrs)" +
theme(plot.title = element_text(hjust = 0.5), plot.subtitle= element_text(hjust = 0.5))

```

king all dU repair components have the most severe fitness defect

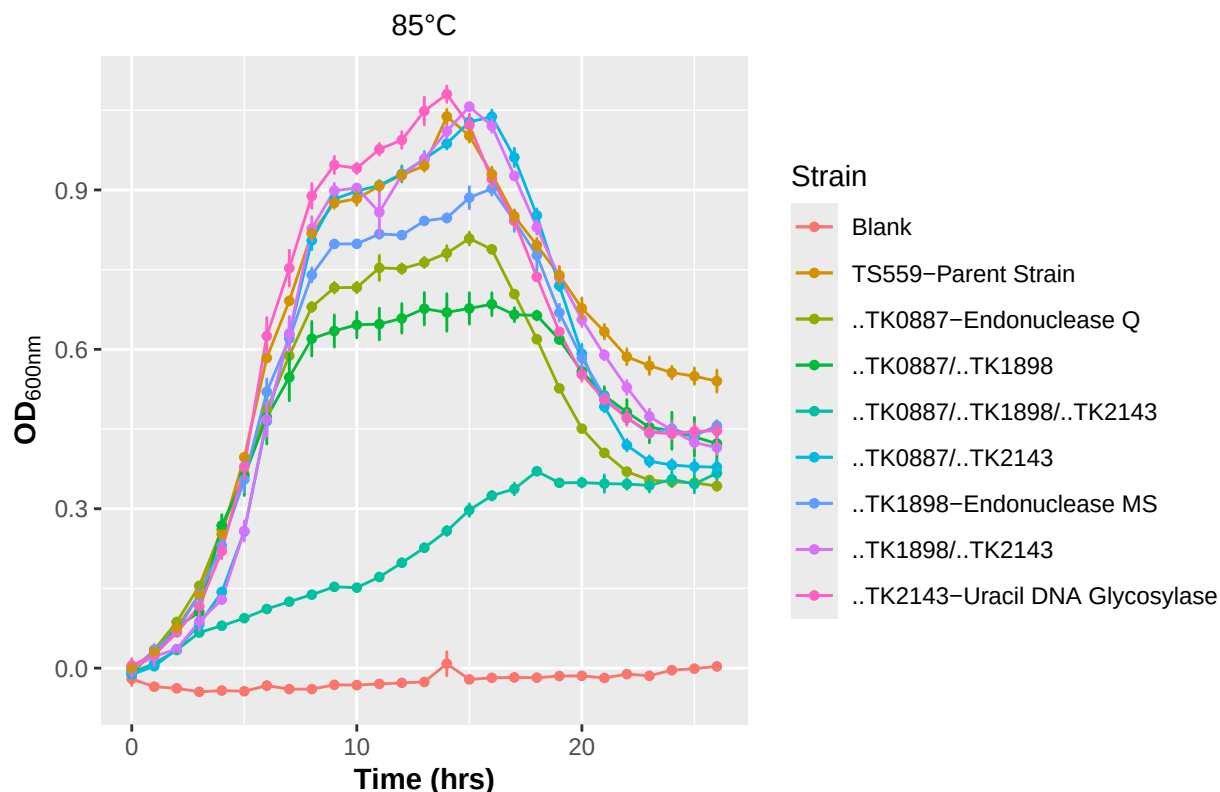


```

dU_GC_tidy_Stat%>%
  ggplot(aes(x=Hours, y=mean, color=Strain, ymax=mean+se, ymin=mean-se)) +
  geom_line(size=0.5) +
  geom_point(size=1.2) +
  geom_errorbar(width=0.1) +
  labs(title = "Strains lacking all dU repair components have the most severe fitness defect", subtitle=
    y = bquote(bold("OD") [600] [nm]),
    x = bquote(bold('Time (hrs)')))) +
  theme(plot.title = element_text(hjust = 0.5), plot.subtitle= element_text(hjust = 0.5))

```

king all dU repair components have the most severe fitness defect



7. Scales

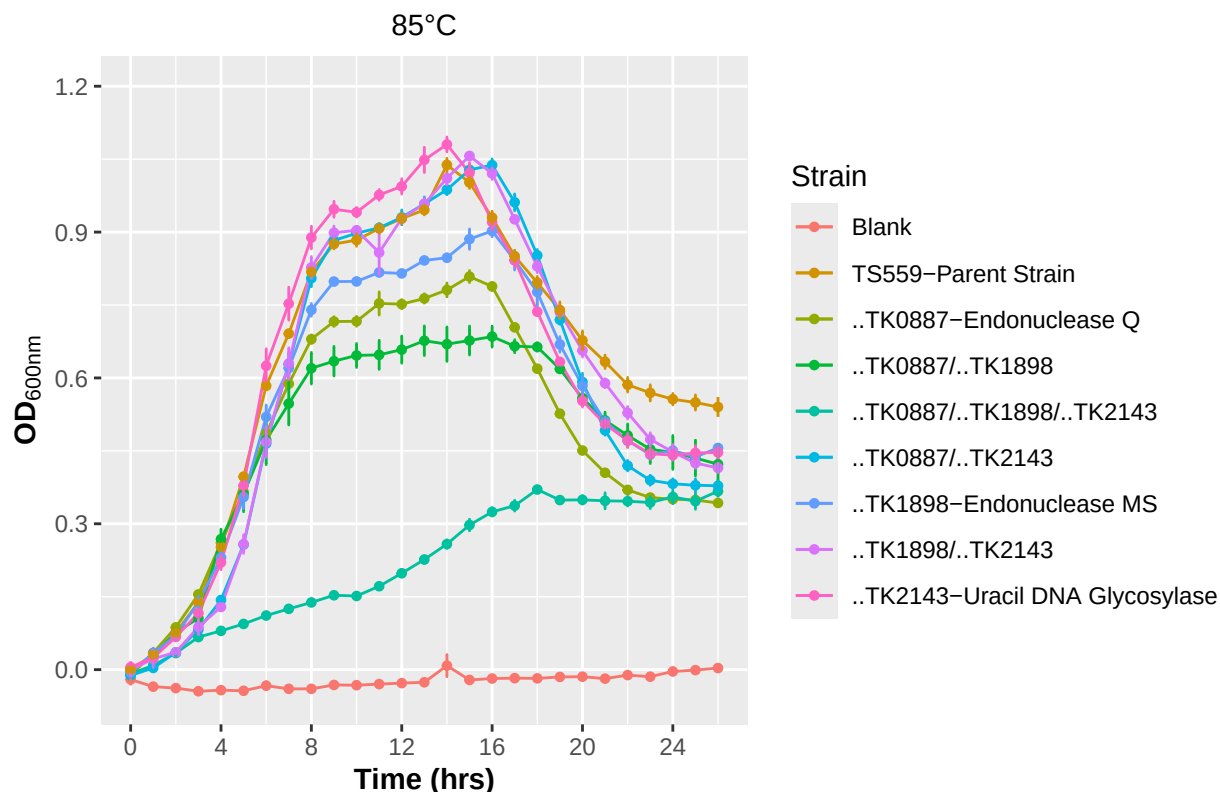
The scales are currently not very informative.

Lets rescale the x axis tick marks to be more helpful. Try to get it to have a tick mark every 4 hours. Try also rescaling the y axis so we have context for where the maximum OD is, try showing OD every 0.3.

```
####
#07#
####

dU_GC_tidy_Stat%>%
  ggplot(aes(x=Hours, y=mean, color=Strain, ymax=mean+se, ymin=mean-se)) +
  geom_line(size=0.5) +
  geom_point(size=1.2) +
  geom_errorbar(width=0.1) +
  labs(title = "Strains lacking all dU repair components have the most severe fitness defect", subtitle=
    y = bquote(bold("OD") [600] [nm]),
    x = bquote(bold('Time (hrs)')))) +
  theme(plot.title = element_text(hjust = 0.5), plot.subtitle= element_text(hjust = 0.5)) +
  scale_x_continuous(breaks=c(0, 4, 8, 12, 16, 20, 24), limits=c(0,26)) +
  scale_y_continuous(breaks=c(0, 0.3, 0.6, 0.9, 1.2), limits=c(-0.05, 1.2))
```

king all dU repair components have the most severe fitness defect



7b. Fitting the Data

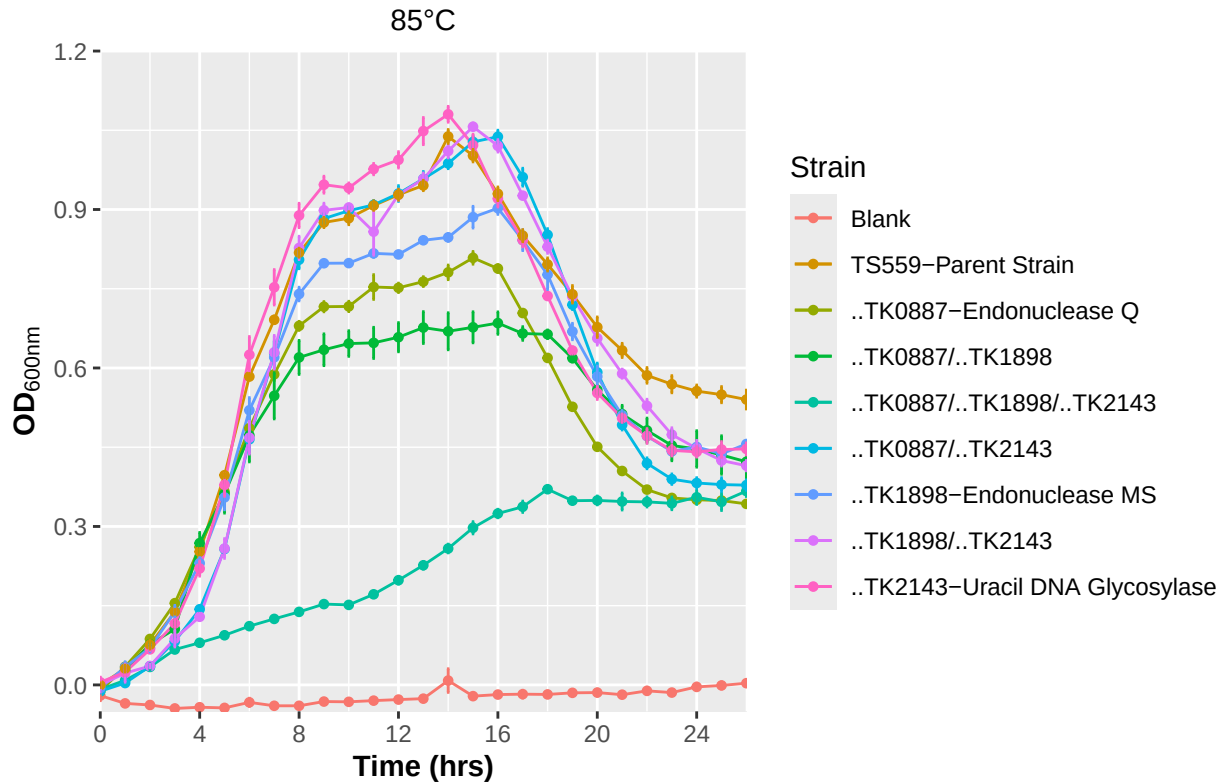
The scales are better...but the data does not start at 0,0. It also goes out past 26 to empty space.

We can add an expand argument to our continuous scales to set the data at the origin.

```
####
#07b#
####

dU_GC_tidy_Stat%>%
  ggplot(aes(x=Hours, y=mean, color=Strain, ymax=mean+se, ymin=mean-se)) +
  geom_line(size=0.5) +
  geom_point(size=1.2) +
  geom_errorbar(width=0.1) +
  labs(title = bquote(bold("Strains lacking all dU repair components have the most severe fitness defect")),
       y = bquote(bold("OD") [600] [nm]),
       x = bquote(bold('Time (hrs)')) +
  theme(plot.title = element_text(hjust = 0.5), plot.subtitle= element_text(hjust = 0.5)) +
  scale_x_continuous(breaks=c(0, 4, 8, 12, 16, 20, 24), limits=c(0,26), expand = c(0,0)) +
  scale_y_continuous(breaks=c(0, 0.3, 0.6, 0.9, 1.2), limits=c(-0.05, 1.2), expand = c(0,0))
```

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7c. Transforming to log scale

That looks good!

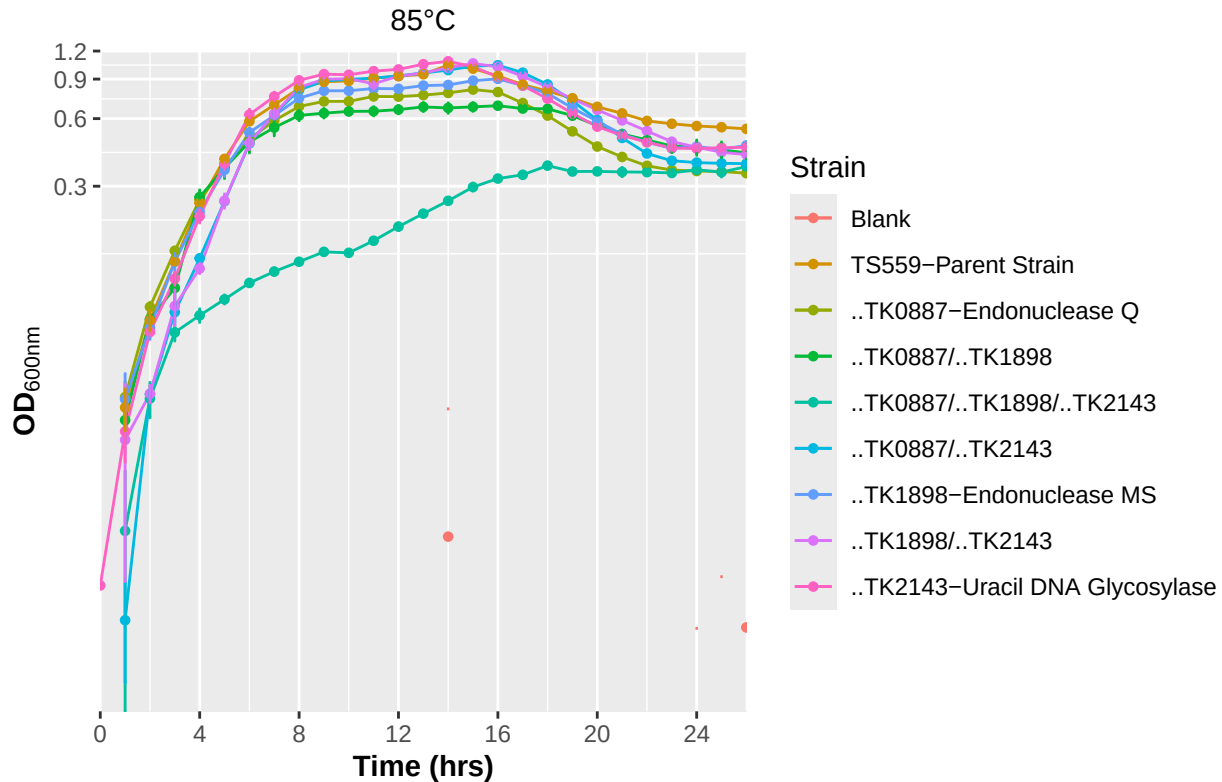
We can also transform the data to be more logarithmic.

NOTE: While log scale is accepted for this data in my field, you should scale your data based on what is appropriate. Notice how all the differences in my graphs are hidden now.

```
####
#07c#
####

dU_GC_tidy_Stat%>%
  ggplot(aes(x=Hours, y=mean, color=Strain, ymax=mean+se, ymin=mean-se)) +
  geom_line(size=0.5) +
  geom_point(size=1.2) +
  geom_errorbar(width=0.1) +
  labs(title = bquote(bold("Strains lacking all dU repair components have the most severe fitness defect")),
       y = bquote(bold("OD") [600] [nm]),
       x = bquote(bold('Time (hrs)')) +
  theme(plot.title = element_text(hjust = 0.5), plot.subtitle= element_text(hjust = 0.5)) +
  scale_x_continuous(breaks=c(0, 4, 8, 12, 16, 20, 24), limits=c(0,26), expand = c(0,0)) +
  scale_y_continuous(breaks=c(0, 0.3, 0.6, 0.9, 1.2), limits=c(-0.05, 1.2), expand = c(0,0), trans = "log")
```

Strains lacking all dU repair components have the most severe fitness defect



8. Scales-Colors

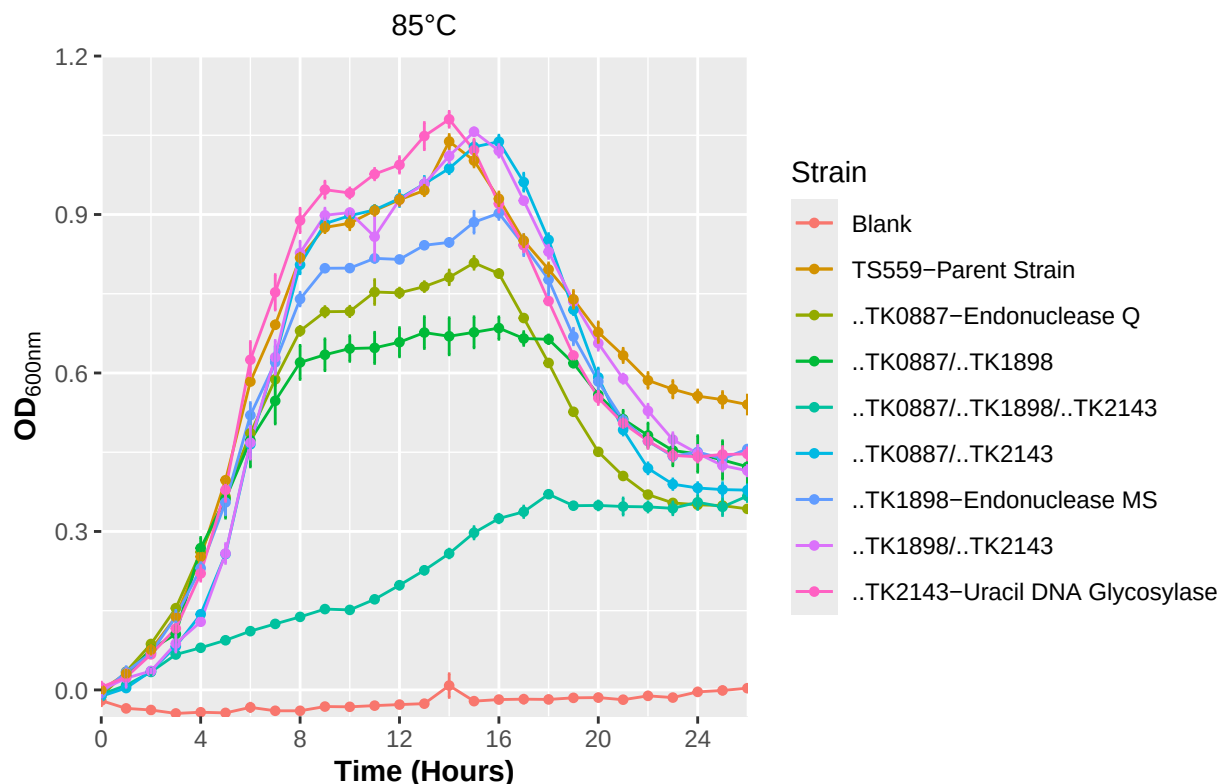
Right now the legend is pretty out of order and the rainbow of colors is hard to follow.

Let's make a color palette we can work with!

```
####
#08#
####

dU_GC_tidy_Stat%>%
  ggplot(aes(x=Hours, y=mean, color=Strain, ymax=mean+se, ymin=mean-se)) +
  geom_line(size=0.5) +
  geom_point(size=1.2) +
  geom_errorbar(width=0.1) +
  labs(title = bquote(bold("Strains lacking all dU repair components have the most severe fitness defect")),
       y = bquote(bold("OD") [600] [nm]),
       x = bquote(bold('Time (Hours)')))) +
  theme(plot.title = element_text(hjust = 0.5), plot.subtitle= element_text(hjust = 0.5)) +
  scale_x_continuous(breaks=c(0, 4, 8, 12, 16, 20, 24), limits=c(0,26), expand = c(0,0)) +
  scale_y_continuous(breaks=c(0, 0.3, 0.6, 0.9, 1.2), limits=c(-0.05, 1.2), expand = c(0,0))
```

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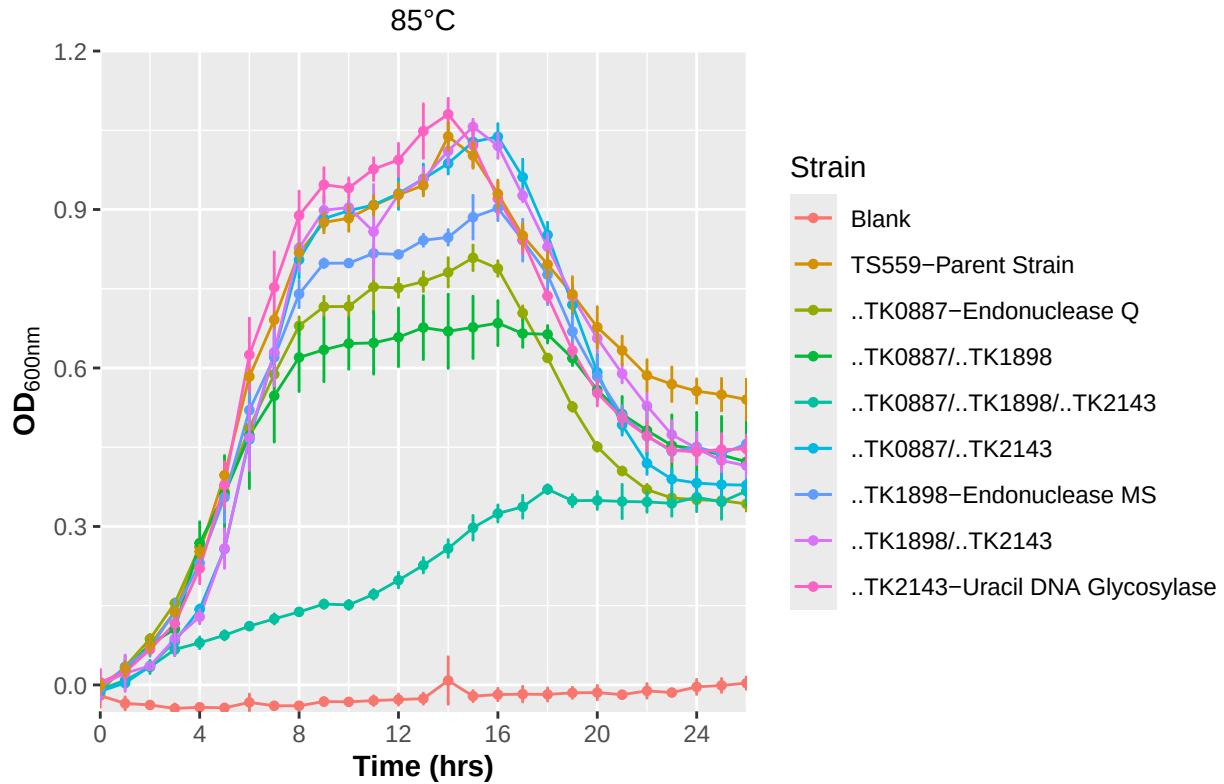
```
scale_color_manual(values=c("pink", "brown4", "red3", "yellow3", "steelblue3", "orange2", "purple3",
```

```
## <ggproto object: Class ScaleDiscrete, Scale, gg>
##   aesthetics: colour
##   axis_order: function
##   break_info: function
##   break_positions: function
##   breaks: waiver
##   call: call
##   clone: function
##   dimension: function
##   drop: TRUE
##   expand: waiver
##   get_breaks: function
##   get_breaks_minor: function
##   get_labels: function
##   get_limits: function
##   get_transformation: function
##   guide: legend
##   is_discrete: function
##   is_empty: function
##   labels: waiver
##   limits: NULL
##   make_sec_title: function
##   make_title: function
```

```
## map: function
## map_df: function
## n.breaks.cache: NULL
## na.translate: TRUE
## na.value: grey50
## name: Strain
## palette: function
## palette.cache: NULL
## position: left
## range: environment
## rescale: function
## reset: function
## train: function
## train_df: function
## transform: function
## transform_df: function
## super: <ggproto object: Class ScaleDiscrete, Scale, gg>
```

```
dU_GC_tidy_Stat%>%
  ggplot() +
  geom_line(aes(x=Hours, y=mean, color=Strain), size=0.5) +
  geom_point(aes(x=Hours, y=mean, color=Strain), size=1.2) +
  geom_errorbar(width=0.1, aes(x=Hours, ymax = mean+sd, ymin = mean-sd, color=Strain)) +
  labs(title = bquote(bold("Strains lacking all dU repair components have the most severe fitness defect")),
        y = bquote(bold("OD") [600] [nm]),
        x = bquote(bold('Time (hrs)')) +
  theme(plot.title = element_text(hjust = 0.5), plot.subtitle= element_text(hjust = 0.5)) +
  scale_x_continuous(breaks=c(0, 4, 8, 12, 16, 20, 24), limits=c(0,26), expand = c(0,0)) +
  scale_y_continuous(breaks=c(0, 0.3, 0.6, 0.9, 1.2), limits=c(-0.05, 1.2), expand = c(0,0))
```

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```
scale_color_brewer(palette="Set3")
```

```
## <ggproto object: Class ScaleDiscrete, Scale, gg>
##   aesthetics: colour
##   axis_order: function
##   break_info: function
##   break_positions: function
##   breaks: waiver
##   call: call
##   clone: function
##   dimension: function
##   drop: TRUE
##   expand: waiver
##   get_breaks: function
##   get_breaks_minor: function
##   get_labels: function
##   get_limits: function
##   get_transformation: function
##   guide: legend
##   is_discrete: function
##   is_empty: function
##   labels: waiver
##   limits: NULL
##   make_sec_title: function
##   make_title: function
```



```
## map: function
## map_df: function
## n.breaks.cache: NULL
## na.translate: TRUE
## na.value: NA
## name: waiver
## palette: function
## palette.cache: NULL
## position: left
## range: environment
## rescale: function
## reset: function
## train: function
## train_df: function
## transform: function
## transform_df: function
## super: <ggproto object: Class ScaleDiscrete, Scale, gg>
```

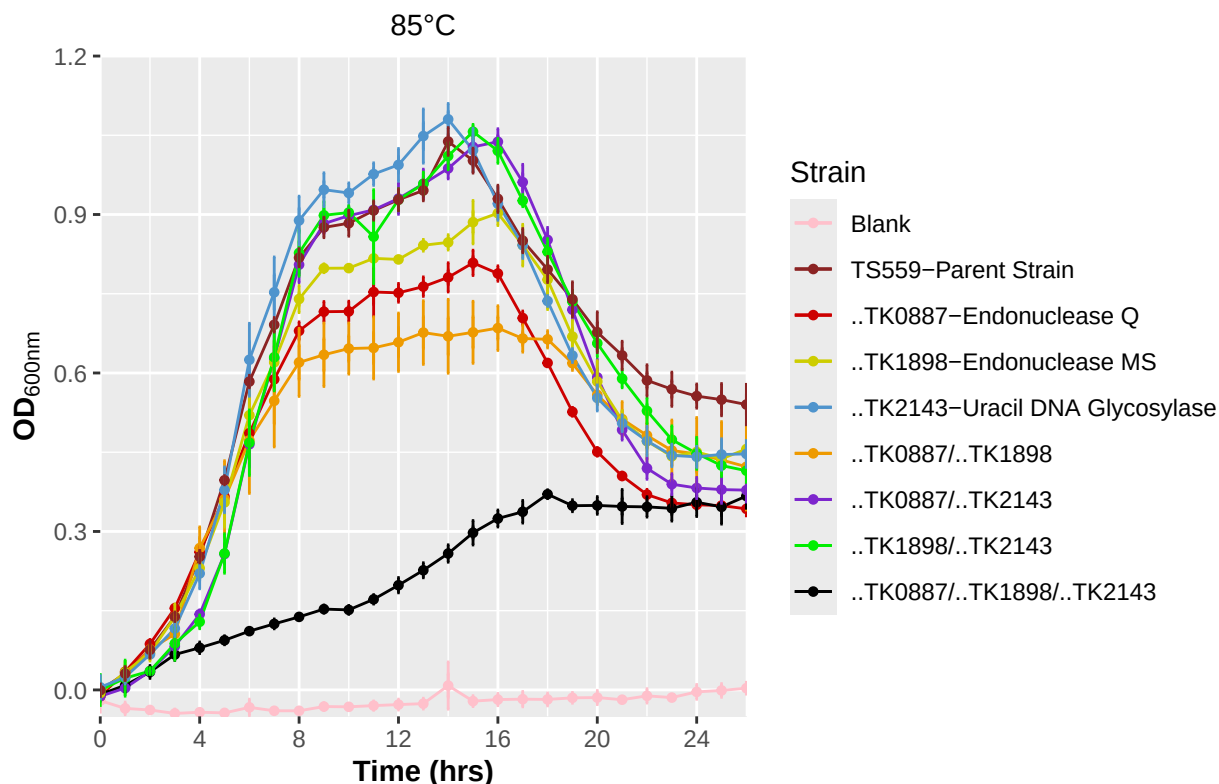
8b. Set your colors to your legend

Now we have our own colors, but the legend is not in any logical order!

As with `scale_x_continuous`, we can also add breaks to `scale_color_manual` to reorder our legend names to our colors.

```
####  
#08b#  
####  
  
dU_GC_tidy_Stat%>%  
  ggplot(aes(x=Hours, y=mean, color=Strain, ymax=mean+se, ymin=mean-se)) +  
  geom_line(size=0.5) +  
  geom_point(size=1.2) +  
  geom_errorbar(width=0.1) +  
  labs(title = bquote(bold("Strains lacking all dU repair components have the most severe fitness defects")),  
        y = bquote(bold("OD")[600][nm]),  
        x = bquote(bold('Time (hrs)')))) +  
  theme(plot.title = element_text(hjust = 0.5), plot.subtitle= element_text(hjust = 0.5)) +  
  scale_x_continuous(breaks=c(0, 4, 8, 12, 16, 20, 24), limits=c(0,26), expand = c(0,0)) +  
  scale_y_continuous(breaks=c(0, 0.3, 0.6, 0.9, 1.2), limits=c(-0.05, 1.2), expand = c(0,0)) +  
  scale_color_manual(values=c("pink", "brown4", "red3", "yellow3", "steelblue3", "orange2", "purple3", "black")) +  
  geom_errorbar(width=0.1, aes(x=Hours, ymax = mean+sd, ymin = mean-sd, color=Strain))
```

ng all dU repair components have the most severe fitness defect



9. Theme

Now we are done graphing, choose your favorite theme.

I prefer minimalism in my axis.

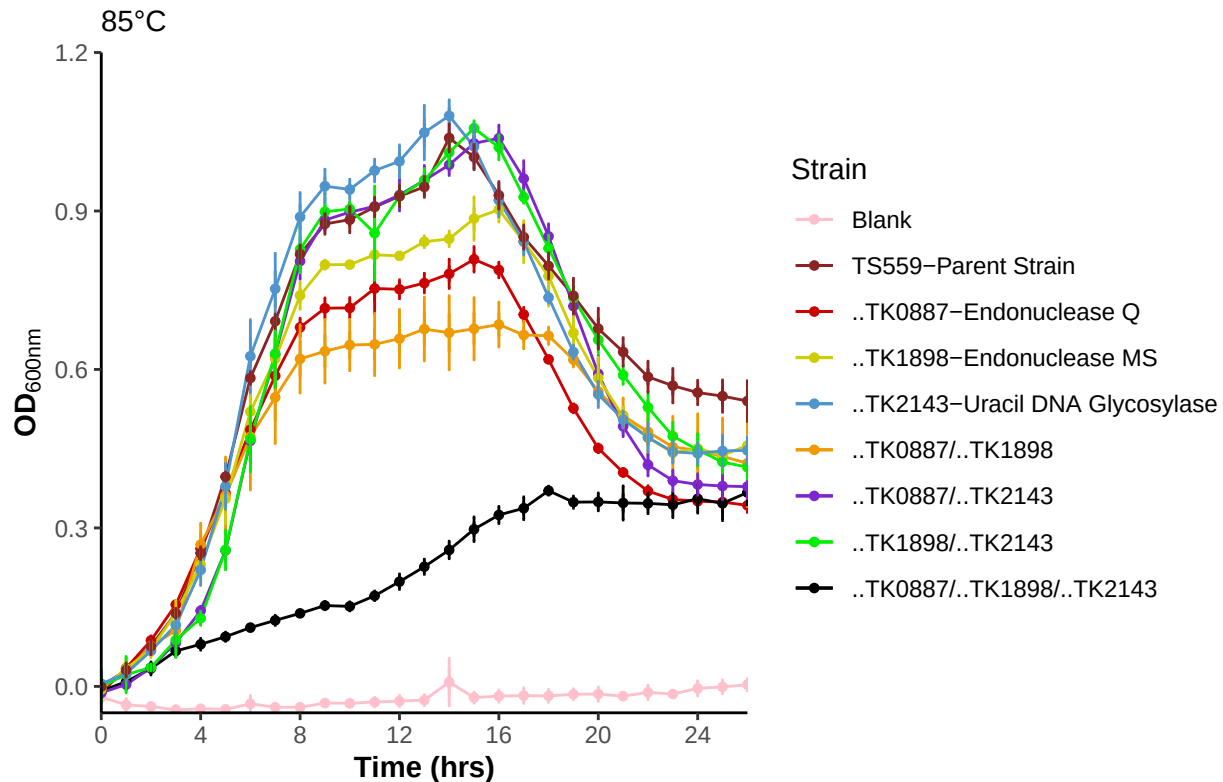
If you put a theme, you will want to put it early in your ggplot code. Otherwise the theme is building off of every previous component and this can cause errors.

```
####
#9#
####

dU_GC_tidy_Stat%>%
  ggplot(aes(x=Hours, y=mean, color=Strain, ymax=mean+se, ymin=mean-se)) +
  geom_line(size=0.5) +
  geom_point(size=1.2) +
  geom_errorbar(width=0.1) +
  labs(title = bquote(bold("Strains lacking all dU repair components have the most severe fitness defect")),
       y = bquote(bold("OD") [600] [nm]),
       x = bquote(bold('Time (hrs)')) +
  theme(plot.title = element_text(hjust = 0.5), plot.subtitle= element_text(hjust = 0.5)) +
  scale_x_continuous(breaks=c(0, 4, 8, 12, 16, 20, 24), limits=c(0,26), expand = c(0,0)) +
  scale_y_continuous(breaks=c(0, 0.3, 0.6, 0.9, 1.2), limits=c(-0.05, 1.2), expand = c(0,0)) +
  scale_color_manual(values=c("pink", "brown4", "red3", "yellow3", "steelblue3", "orange2", "purple3",
```

```
geom_errorbar(width=0.1, aes(x=Hours, ymax = mean+sd, ymin = mean-sd, color=Strain)) +
theme_classic()
```

Strains lacking all dU repair components have the most severe fitness



9b. Adjusting Themes

But what if I wanted a different theme and wanted no axis?

Watch out-different themes will resize your axis, title, and legend, so you will have to fix this in post.

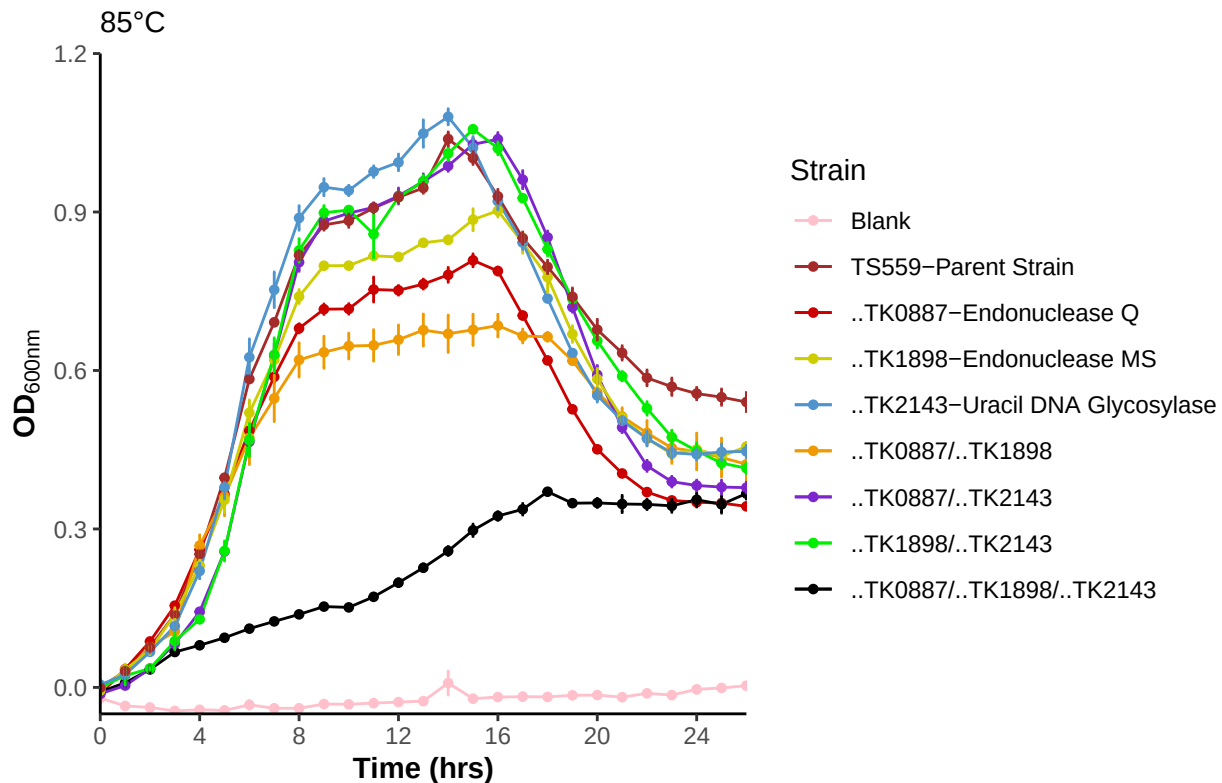
```
####
#9b#
####
```

#using a theme

```
dU_GC_tidy_Stat%>%
  ggplot(aes(x=Hours, y=mean, color=Strain, ymax=mean+se, ymin=mean-se)) +
  geom_line(size=0.5) +
  geom_point(size=1.2) +
  geom_errorbar(width=0.1) +
  labs(title = bquote(bold("Strains lacking all dU repair components have the most severe fitness defect")),
       y = bquote(bold("OD") [600] [nm]),
       x = bquote(bold('Time (hrs)')))) +
```

```
theme(plot.title = element_text(hjust = 0.5), plot.subtitle= element_text(hjust = 0.5)) +
scale_x_continuous(breaks=c(0, 4, 8, 12, 16, 20, 24), limits=c(0,26), expand = c(0,0)) +
scale_y_continuous(breaks=c(0, 0.3, 0.6, 0.9, 1.2), limits=c(-0.05, 1.2), expand = c(0,0)) +
scale_color_manual(values=c("pink", "brown", "red3", "yellow3", "steelblue3", "orange2", "purple3", "black")) +
theme(panel.grid.major = element_blank(), panel.grid.minor = element_blank()) +
theme_classic()
```

Strains lacking all dU repair components have the most severe fitness

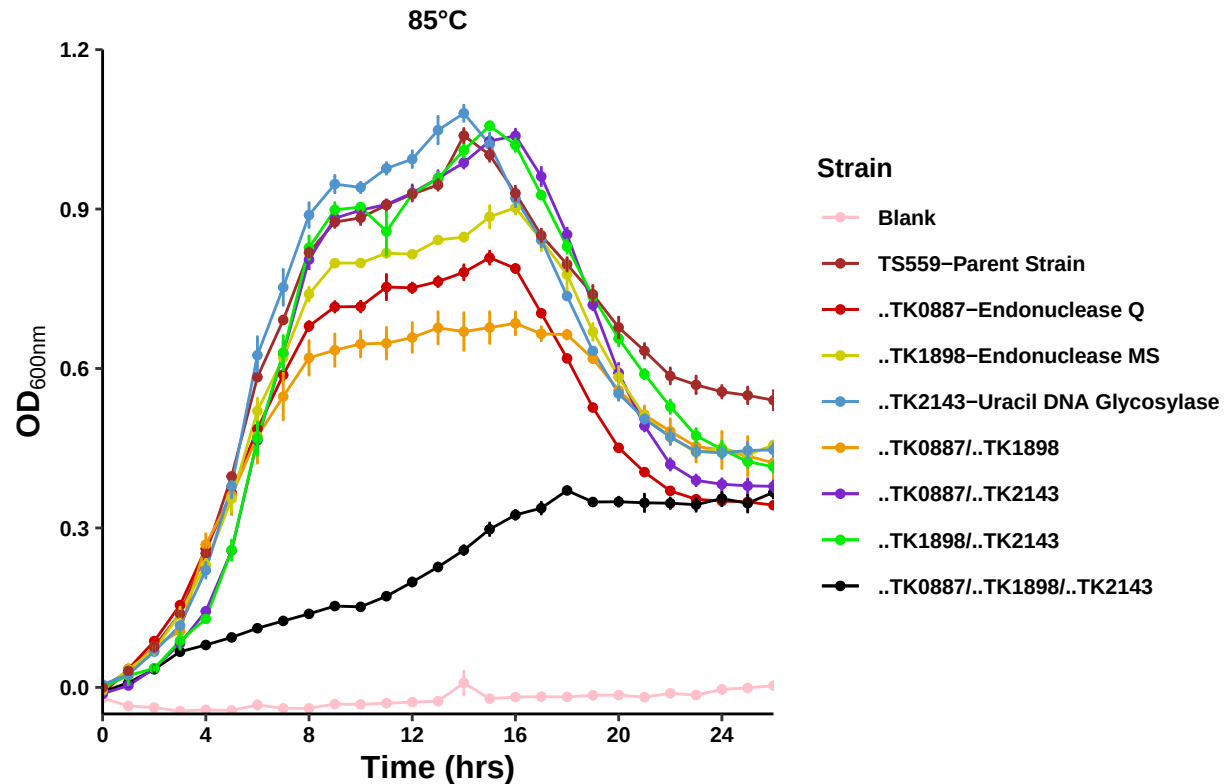


#using and adjusting the theme

```
dU_GC_tidy_Stat%>%
ggplot(aes(x=Hours, y=mean, color=Strain, ymax=mean+se, ymin=mean-se)) +
geom_line(size=0.5) +
geom_point(size=1.2) +
geom_errorbar(width=0.1) +
labs(title = bquote(bold("Strains lacking all dU repair components have the most severe fitness defect")),
      y = bquote(bold("OD") [600] [nm]),
      x = bquote(bold('Time (hrs)')) +
scale_x_continuous(breaks=c(0, 4, 8, 12, 16, 20, 24), limits=c(0,26), expand = c(0,0)) +
scale_y_continuous(breaks=c(0, 0.3, 0.6, 0.9, 1.2), limits=c(-0.05, 1.2), expand = c(0,0)) +
scale_color_manual(values=c("pink", "brown", "red3", "yellow3", "steelblue3", "orange2", "purple3", "black")) +
theme_classic() +
theme(plot.title = element_text(hjust = 0.5, size = 12), plot.subtitle= element_text(hjust = 0.5, size = 12),
      theme(axis.text.x = element_text(face = 'bold', size = 8, color = "black")) +
      theme(axis.text.y = element_text(face = 'bold', size = 8, color = 'black')) +
      theme(axis.title.x = element_text(face = 'bold', size = 12)) +
```

```
theme(axis.title.y = element_text(face = 'bold', size = 12)) +
theme(panel.grid.major = element_blank(), panel.grid.minor = element_blank()) +
theme(legend.text = element_text(face = 'bold', size = 8)) +
theme(legend.title = element_text(face = 'bold', size = 10))
```

cking all dU repair components have the most severe fitness defect



10. ggSave

It looks like we are ready to save the graph!

Let's use ggsave.

```
####
#10#
####

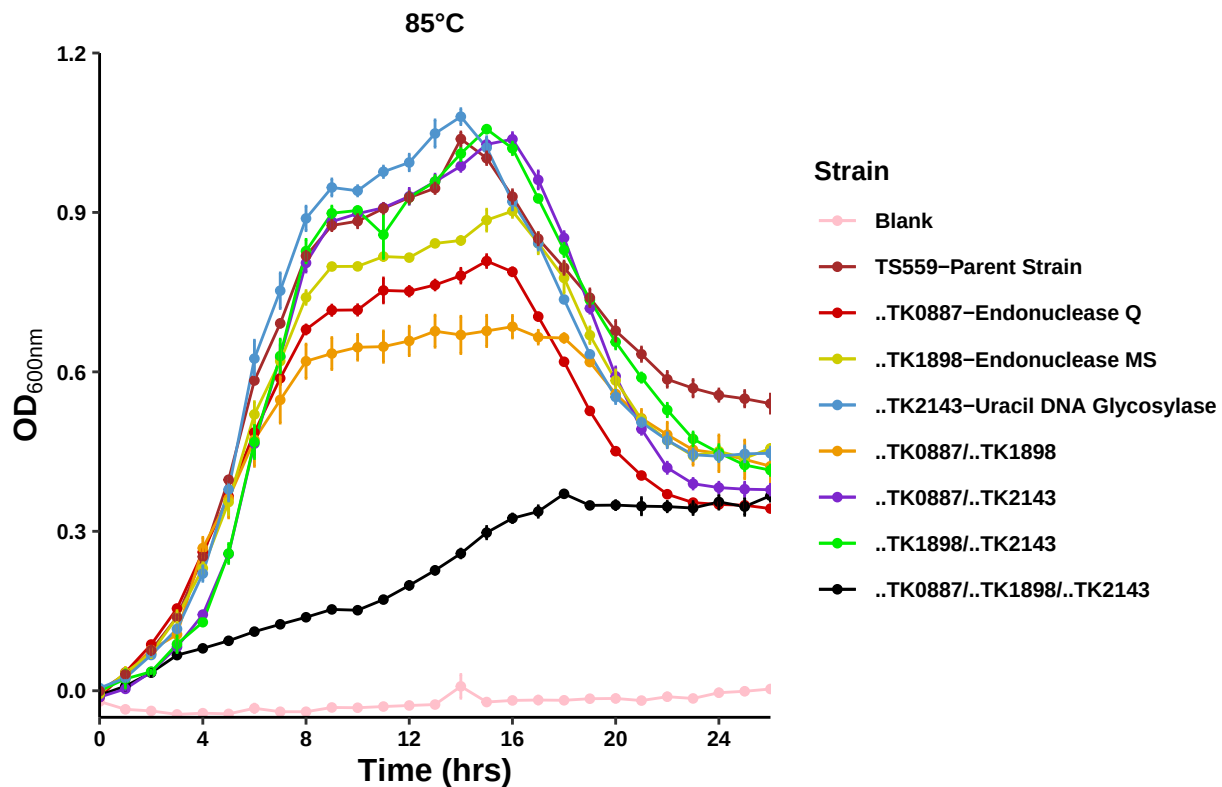
dU_GC_tidy_Stat%>%
  ggplot(aes(x=Hours, y=mean, color=Strain, ymax=mean+se, ymin=mean-se)) +
  geom_line(size=0.5) +
  geom_point(size=1.2) +
  geom_errorbar(width=0.1) +
  labs(title = bquote(bold("Strains lacking all dU repair components have the most severe fitness defect")),
       y = bquote(bold("OD") [600] [nm]),
       x = bquote(bold("Time (hrs)"))) +
  scale_x_continuous(breaks=c(0, 4, 8, 12, 16, 20, 24), limits=c(0,26), expand = c(0,0)) +
```

```

scale_y_continuous(breaks=c(0, 0.3, 0.6, 0.9, 1.2), limits=c(-0.05, 1.2), expand = c(0,0)) +
scale_color_manual(values=c("pink", "brown", "red3", "yellow3", "steelblue3", "orange2", "purple3", "black")) +
theme_classic() +
theme(plot.title = element_text(hjust = 0.5, size = 12), plot.subtitle= element_text(hjust = 0.5, size = 12)) +
theme(axis.text.x = element_text(face = 'bold', size = 8, color = "black")) +
theme(axis.text.y = element_text(face = 'bold', size = 8, color = 'black')) +
theme(axis.title.x = element_text(face = 'bold', size = 12)) +
theme(axis.title.y = element_text(face = 'bold', size = 12)) +
theme(panel.grid.major = element_blank(), panel.grid.minor = element_blank()) +
theme(legend.text = element_text(face = 'bold', size = 8)) +
theme(legend.title = element_text(face = 'bold', size = 10))

```

cking all dU repair components have the most severe fitness defect



```

ggsave("dU Growth Curve.jpg", dpi=2000, width = 8.25, height = 4, unit = 'in')

```

11. Facet Wrapping

BUT WAIT! THERE IS MORE!

We never did figure out the issue of overlapping graphs after all.

We can separate the graphs into smaller individual graphs for each strain.

```

####
#11#

```

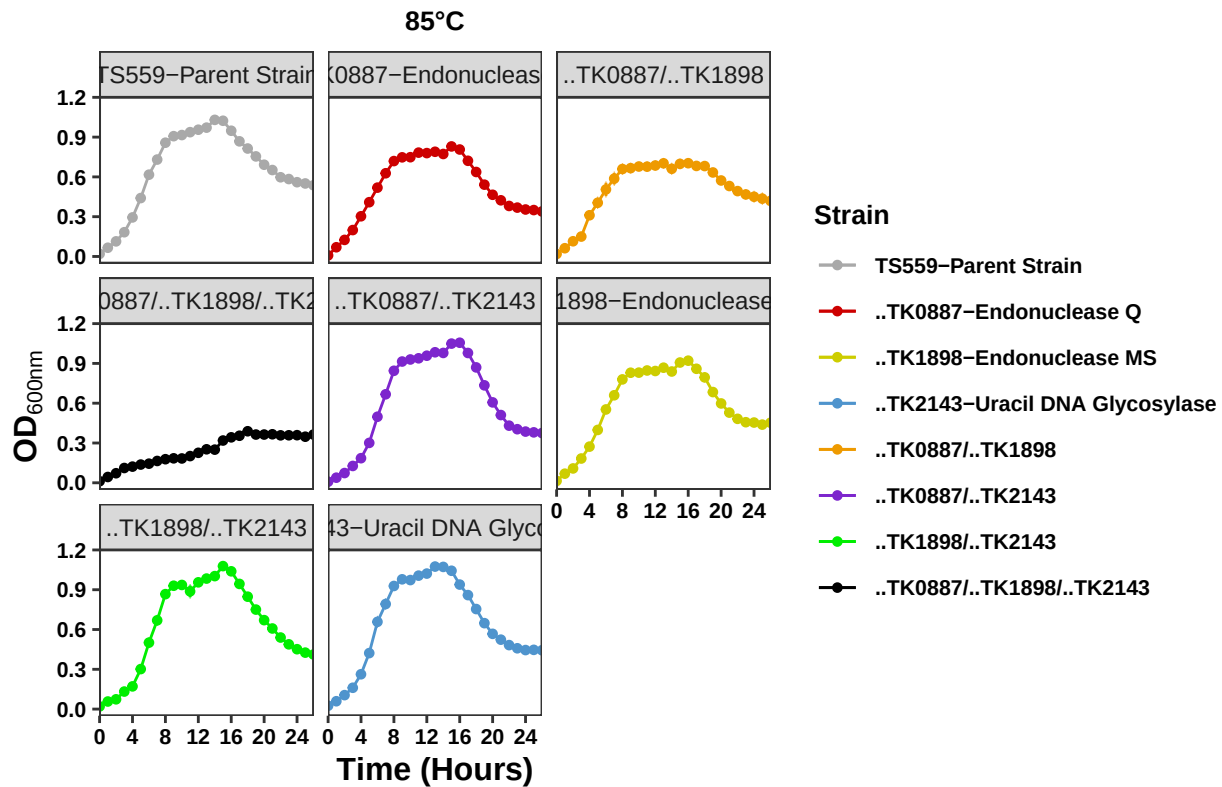
```
####
dU_GC_tidy<-read.csv("Growth_Curve_dU_tidy.csv",fill=TRUE,header=TRUE)

dU_Neg_85 <- dU_GC_tidy %>%
  filter(Strain == "Blank") %>%
  group_by(Hours)%>%
  summarize(blank_mean = mean(OD), .groups = "drop")

dU_GC_tidy_Stat<- dU_GC_tidy %>%
  left_join(dU_Neg_85, by = "Hours") %>%
  mutate(OD = OD - blank_mean) %>%
  group_by(Hours, Strain) %>%
  summarize(mean = mean(OD), sd = sd(OD), n = n(), se = sd / sqrt(n), .groups = "drop") %>%
  filter(Strain != "Blank") %>%
  mutate(Strain =
    recode(Strain, "TK0887-Endonuclease Q" = "\u0394TK0887-Endonuclease Q",
      "TK1898-Endonuclease MS" = "\u0394TK1898-Endonuclease MS",
      "TK2143-Uracil DNA Glycosylase" = "\u0394TK2143-Uracil DNA Glycosylase",
      "TK0887/TK1898" = "\u0394TK0887/\u0394TK1898",
      "TK0887/TK2143" = "\u0394TK0887/\u0394TK2143",
      "TK1898/TK2143" = "\u0394TK1898/\u0394TK2143",
      "TK0887/TK1898/TK2143" = "\u0394TK0887/\u0394TK1898/\u0394TK2143"))

dU_GC_tidy_Stat%>%
  ggplot(aes(x=Hours, y=mean, color=Strain, ymax=mean+se, ymin=mean-se)) +
  geom_line(size=0.5) +
  geom_point(size=1.2) +
  geom_errorbar(width=0.1) +
  theme_bw() +
  labs(title = bquote(bold("Strains lacking all dU repair components have the most severe fitness defect")),
    y = bquote(bold("OD") [600] [nm]),
    x = bquote(bold('Time (Hours)')))) +
  scale_x_continuous(breaks=c(0, 4, 8, 12, 16, 20, 24), limits=c(0,26), expand = c(0,0)) +
  scale_y_continuous(breaks=c(0, 0.3, 0.6, 0.9, 1.2), limits=c(-0.05, 1.2), expand = c(0,0)) +
  scale_color_manual(values=c( "darkgrey", "red3", "yellow3", "steelblue3", "orange2", "purple3", "green3")) +
  facet_wrap(~Strain) +
  theme(panel.grid.major = element_blank(), panel.grid.minor = element_blank()) +
  theme(plot.title = element_text(hjust = 0.5, size = 12), plot.subtitle= element_text(hjust = 0.5, size = 12)) +
  theme(axis.text.x = element_text(face = 'bold', size = 8, color = "black")) +
  theme(axis.text.y = element_text(face = 'bold', size = 8, color = 'black')) +
  theme(axis.title.x = element_text(face = 'bold', size = 12)) +
  theme(axis.title.y = element_text(face = 'bold', size = 12)) +
  theme(panel.grid.major = element_blank(), panel.grid.minor = element_blank()) +
  theme(legend.text = element_text(face = 'bold', size = 8)) +
  theme(legend.title = element_text( face = 'bold', size = 10))
```

cking all dU repair components have the most severe fitness defect



```
ggsave("dU_Plot_85_Facet_Wrapped.jpg", dpi=2000)
```

```
#facet_wrap(~factor(Strain, levels=c( "TS559-Parent Strain", "\u0394TK0887-Endonuclease Q", "\u0394TK1898-Endonuclease MS", "\u0394TK2143-Uracil DNA Glycosylase", "\u0394TK0887/TK1898/TK2143" )))
```

11b. Facet-Wrap - Removing Legends

It is a bit redundant to have the panel labels and the strain labels.

We can remove one or the other.

```
####
#11b#
####

#without the legend

dU_GC_tidy_Stat%>%
  ggplot(aes(x=Hours, y=mean, color=Strain, ymax=mean+se, ymin=mean-se)) +
  geom_line(size=0.5) +
  geom_point(size=1.2) +
  geom_errorbar(width=0.1) +
  theme_bw() +
```

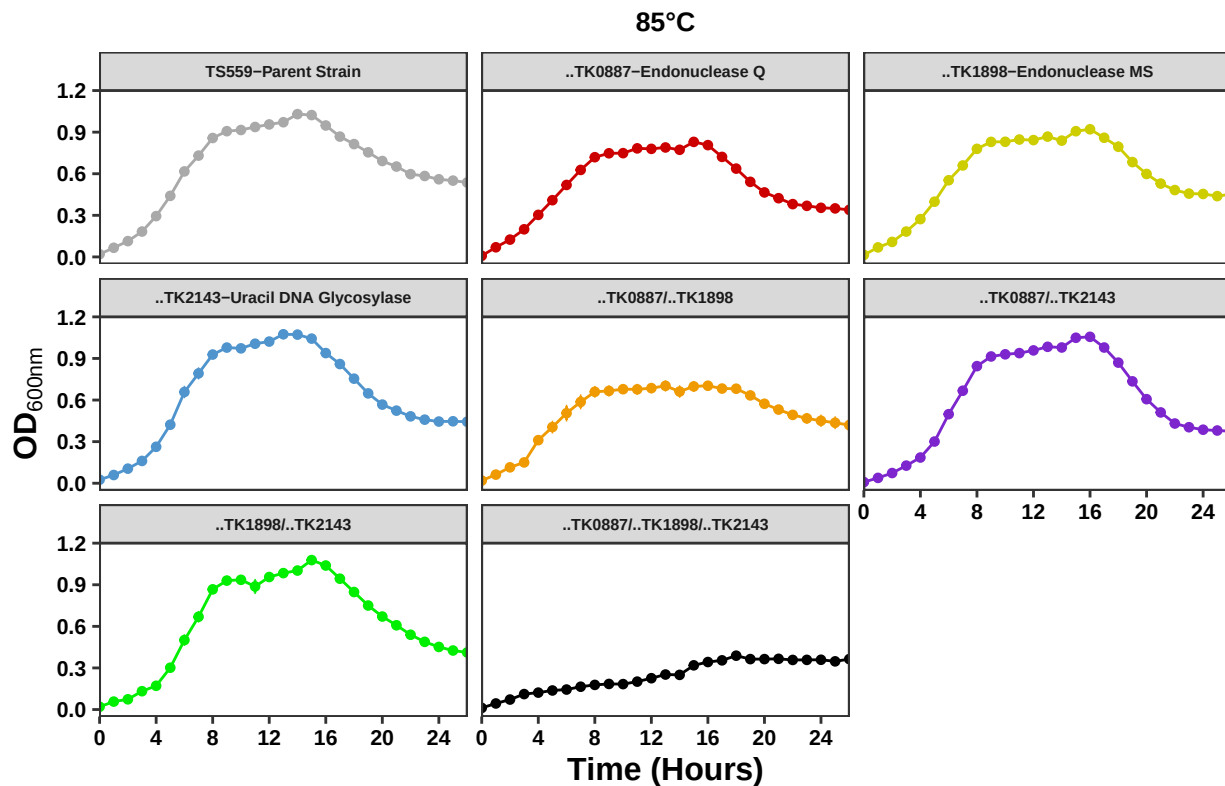


```

labs(title = bquote(bold("Strains lacking all dU repair components have the most severe fitness defect")),
      y = bquote(bold("OD") [600] [nm]),
      x = bquote(bold('Time (Hours)')))) +
scale_x_continuous(breaks=c(0, 4, 8, 12, 16, 20, 24), limits=c(0,26), expand = c(0,0)) +
scale_y_continuous(breaks=c(0, 0.3, 0.6, 0.9, 1.2), limits=c(-0.05, 1.2), expand = c(0,0)) +
scale_color_manual(values=c( "darkgrey", "red3", "yellow3", "steelblue3", "orange2", "purple3", "green3", "black")) +
facet_wrap(~factor(Strain, levels=c( "TS559-Parent Strain", "\u0394TK0887-Endonuclease Q", "\u0394TK1898-Endonuclease MS",
                                     "\u0394TK2143-Uracil DNA Glycosylase",
                                     "\u0394TK0887/\u0394TK1898",
                                     "\u0394TK0887/\u0394TK2143",
                                     "\u0394TK1898/\u0394TK2143",
                                     "\u0394TK0887/\u0394TK1898/\u0394TK2143"))) +
  theme(panel.grid.major = element_blank(), panel.grid.minor = element_blank()) +
  theme(plot.title = element_text(hjust = 0.5, size = 12), plot.subtitle= element_text(hjust = 0.5, size = 12)) +
  theme(axis.text.x = element_text(face = 'bold', size = 8, color = "black")) +
  theme(axis.text.y = element_text(face = 'bold', size = 8, color = 'black')) +
  theme(axis.title.x = element_text(face = 'bold', size = 12)) +
  theme(axis.title.y = element_text(face = 'bold', size = 12)) +
  theme(panel.grid.major = element_blank(), panel.grid.minor = element_blank()) +
  theme(legend.text = element_text(face = 'bold', size = 8)) +
  theme(legend.title = element_text( face = 'bold', size = 10)) +
  theme(strip.text = element_text(face = "bold", size=6))+
  theme(legend.position = "none")

```

Strains lacking all dU repair components have the most severe fitness defect

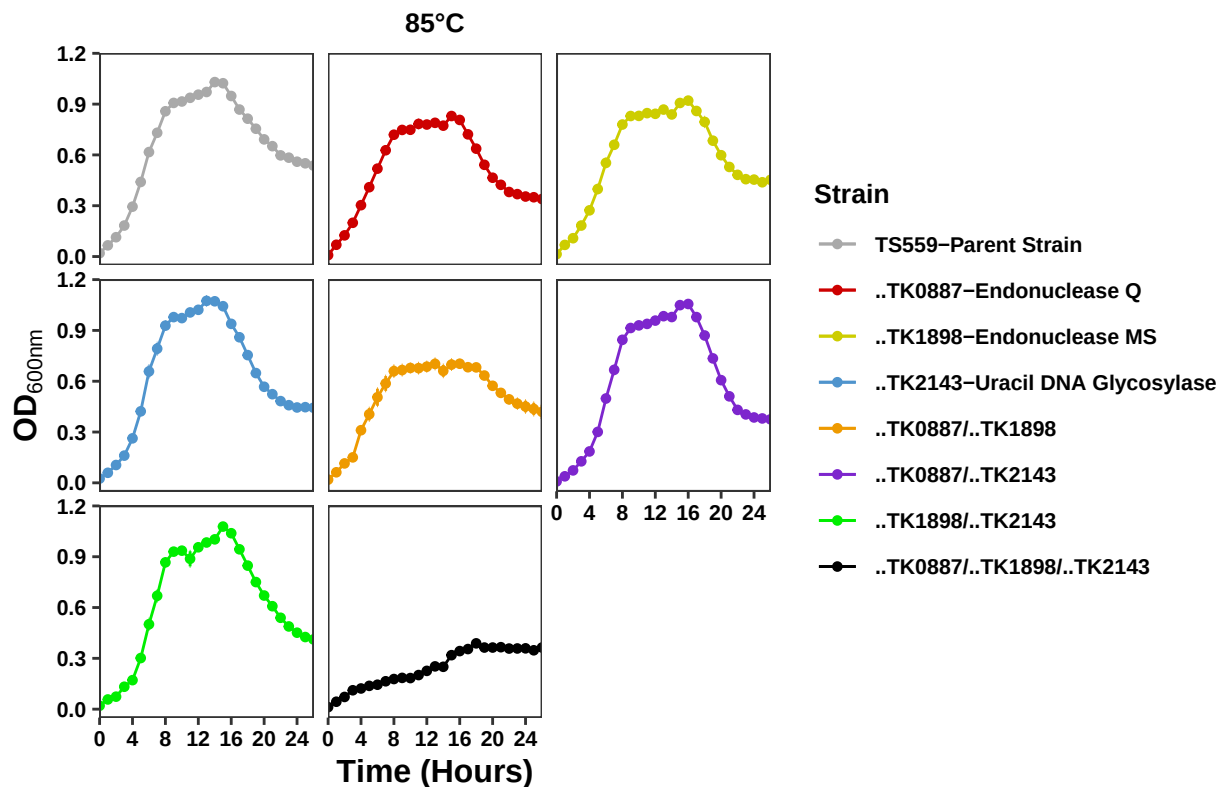


```
ggsave("dU Growth Curve_Facet_Wrapped.jpg", dpi = 2000)
```

```
#without the panel labels
```

```
dU_GC_tidy_Stat%>%
  ggplot(aes(x=Hours, y=mean, color=Strain, ymax=mean+se, ymin=mean-se)) +
  geom_line(size=0.5) +
  geom_point(size=1.2) +
  geom_errorbar(width=0.1) +
  theme_bw() +
  labs(title = bquote(bold("Strains lacking all dU repair components have the most severe fitness defect")),
       y = bquote(bold("OD") [600] [nm]),
       x = bquote(bold('Time (Hours)')))) +
  scale_x_continuous(breaks=c(0, 4, 8, 12, 16, 20, 24), limits=c(0,26), expand = c(0,0)) +
  scale_y_continuous(breaks=c(0, 0.3, 0.6, 0.9, 1.2), limits=c(-0.05, 1.2), expand = c(0,0)) +
  scale_color_manual(values=c( "darkgrey", "red3", "yellow3", "steelblue3", "orange2", "purple3", "green3", "black")) +
  facet_wrap(~factor(Strain, levels=c( "TS559-Parent Strain", "\u00394TK0887-Endonuclease Q", "\u00394TK1898-Endonuclease MS", "\u00394TK2143-Uracil DNA Glycosylase", "\u00394TK0887\u002F\u00394TK1898", "\u00394TK0887\u002F\u00394TK2143", "\u00394TK1898\u002F\u00394TK2143", "\u00394TK0887\u002F\u00394TK1898\u002F\u00394TK2143"))) +
  theme(panel.grid.major = element_blank(), panel.grid.minor = element_blank()) +
  theme(plot.title = element_text(hjust = 0.5, size = 12), plot.subtitle = element_text(hjust = 0.5, size = 12)) +
  theme(axis.text.x = element_text(face = 'bold', size = 8, color = "black")) +
  theme(axis.text.y = element_text(face = 'bold', size = 8, color = 'black')) +
  theme(axis.title.x = element_text(face = 'bold', size = 12)) +
  theme(axis.title.y = element_text(face = 'bold', size = 12)) +
  theme(panel.grid.major = element_blank(), panel.grid.minor = element_blank()) +
  theme(legend.text = element_text(face = 'bold', size = 8)) +
  theme(legend.title = element_text( face = 'bold', size = 10)) +
  theme( strip.text.x = element_blank() )
```

cking all dU repair components have the most severe fitness defect



```
ggsave("dU Growth Curve_Facet_Wrapped_2.jpg", dpi=2000, width = 12)
```

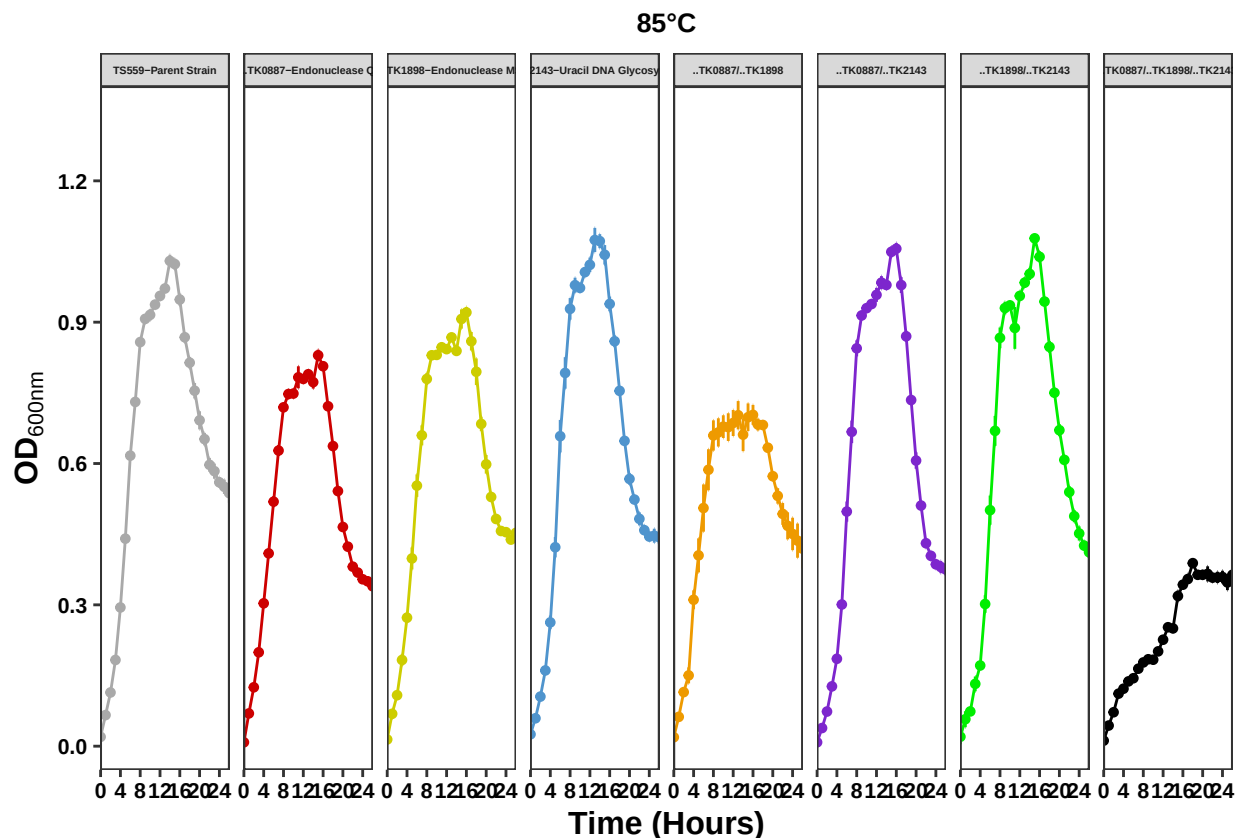
12. Bonus- ggpubr

There are a TON of packages that can take ggplot to the next level.

I actually collected data for both 85 degrees and 95 degrees C.

If I want, I can use the ggpubr package to get an image with two graphs.

```
####  
#12#  
####  
  
#make 85 degree data into an item  
  
dU_85 <- dU_GC_tidy_Stat%>%  
  ggplot(aes(x=Hours, y=mean, color=Strain, ymax=mean+se, ymin=mean-se)) +  
  geom_line(size=0.5) +  
  geom_point(size=1.2) +  
  geom_errorbar(width=0.1) +  
  theme_bw() +  
  labs(subtitle=bquote(bold("85\u00B0C")),  
        y = bquote(bold("OD") [600] [nm]),  
        x = bquote(bold('Time (Hours)')))) +  
  scale_x_continuous(breaks=c(0, 4, 8, 12, 16, 20, 24), limits=c(0,26), expand = c(0,0)) +  
  scale_y_continuous(breaks=c(0, 0.3, 0.6, 0.9, 1.2), limits=c(-0.05, 1.4), expand = c(0,0)) +  
  scale_color_manual(values=c( "darkgrey", "red3", "yellow3", "steelblue3", "orange2", "purple3", "green3", "black")) +  
  facet_wrap(ncol=8, ~factor(Strain, levels=c( "TS559-Parent Strain", "\u0394TK0887-Endonuclease Q", "\u0394TK0887-Endonuclease Q", "\u0394TK0887-Endonuclease Q", "\u0394TK0887-Endonuclease Q", "\u0394TK0887-Endonuclease Q", "\u0394TK0887-Endonuclease Q", "\u0394TK0887-Endonuclease Q"))) +  
  theme(panel.grid.major = element_blank(), panel.grid.minor = element_blank()) +  
  theme(plot.title = element_text(hjust = 0.5, size = 12), plot.subtitle= element_text(hjust = 0.5, size = 12)) +  
  theme(axis.text.x = element_text(face = 'bold', size = 8, color = "black")) +  
  theme(axis.text.y = element_text(face = 'bold', size = 8, color = 'black')) +  
  theme(axis.title.x = element_text(face = 'bold', size = 12)) +  
  theme(axis.title.y = element_text(face = 'bold', size = 12)) +  
  theme(panel.grid.major = element_blank(), panel.grid.minor = element_blank()) +  
  theme(legend.text = element_text(face = 'bold', size = 8)) +  
  theme(legend.title = element_text( face = 'bold', size = 10)) +  
  theme(strip.text = element_text(face = "bold", size=4))+  
  theme(legend.position = "none")  
  
dU_85
```



#95 degree data

```
dU_GC_tidy_95<-read.csv("Growth_Curve_dU_tidy_95.csv",fill=TRUE,header=TRUE)
```

```
dU_Neg_95 <- dU_GC_tidy_95 %>%
  filter(Strain == "Blank") %>%
  group_by(Hours)%>%
  summarize(blank_mean = mean(OD), .groups = "drop")
```

```
dU_GC_tidy_Stat_95<- dU_GC_tidy_95 %>%
  left_join(dU_Neg_95, by = "Hours") %>%
  mutate(OD = OD - blank_mean) %>%
  group_by(Hours, Strain) %>%
  summarize(mean = mean(OD), sd = sd(OD), n = n(), se = sd / sqrt(n), .groups = "drop") %>%
  filter(Strain != "Blank") %>%
  mutate(Strain =
    recode(Strain, "TK0887-Endonuclease Q" = "\u0394TK0887-Endonuclease Q",
      "TK1898-Endonuclease MS" = "\u0394TK1898-Endonuclease MS",
      "TK2143-Uracil DNA Glycosylase" = "\u0394TK2143-Uracil DNA Glycosylase",
      "TK0887/TK1898" = "\u0394TK0887/\u0394TK1898",
      "TK0887/TK2143" = "\u0394TK0887/\u0394TK2143",
      "TK1898/TK2143" = "\u0394TK1898/\u0394TK2143",
      "TK0887/TK1898/TK2143" = "\u0394TK0887/\u0394TK1898/\u0394TK2143"))
```

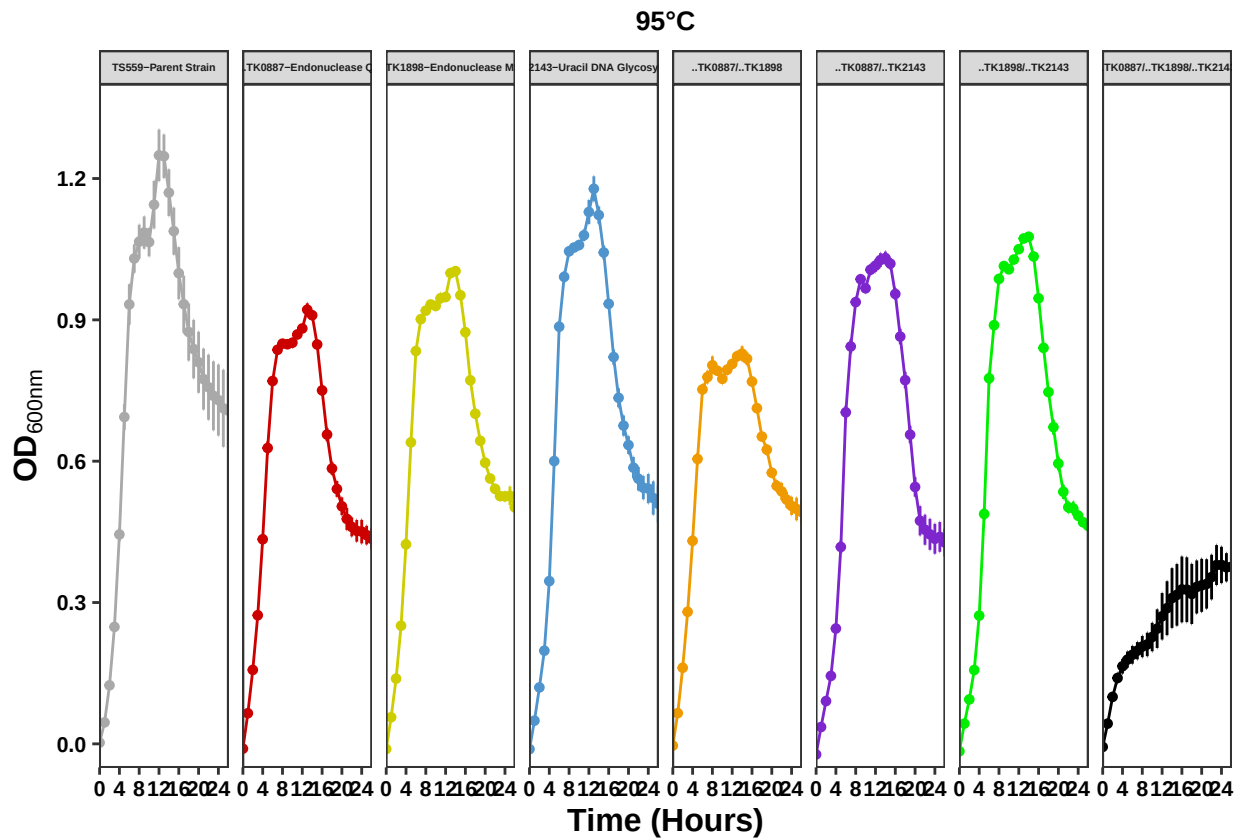
```
dU_95 <- dU_GC_tidy_Stat_95%>%
  ggplot(aes(x=Hours, y=mean, color=Strain, ymax=mean+se, ymin=mean-se)) +
```

```

geom_line(size=0.5) +
geom_point(size=1.2) +
geom_errorbar(width=0.1) +
theme_bw() +
labs( subtitle=bquote(bold("95\u00B0C")),
      y = bquote(bold("OD" [600] [nm])),
      x = bquote(bold('Time (Hours)')) ) +
scale_x_continuous(breaks=c(0, 4, 8, 12, 16, 20, 24), limits=c(0,26), expand = c(0,0)) +
scale_y_continuous(breaks=c(0, 0.3, 0.6, 0.9, 1.2), limits=c(-0.05, 1.4), expand = c(0,0)) +
scale_color_manual(values=c( "darkgrey", "red3", "yellow3", "steelblue3", "orange2", "purple3", "green3", "black" ))
facet_wrap(ncol=8, ~factor(Strain, levels=c( "TS559-Parent Strain", "\u0394TK0887-Endonuclease Q", "\u0394TK1898-Endonuclease M", "\u0394TK2143-Endonuclease Q", "\u0394TK1898\u0394TK2143-Endonuclease Q", "\u0394TK0887\u0394TK1898-Endonuclease Q", "\u0394TK0887\u0394TK2143-Endonuclease Q", "\u0394TK1898\u0394TK0887\u0394TK2143-Endonuclease Q" )))
theme(panel.grid.major = element_blank(), panel.grid.minor = element_blank()) +
theme(plot.title = element_text(hjust = 0.5, size = 12), plot.subtitle= element_text(hjust = 0.5, size = 12)) +
theme(axis.text.x = element_text(face = 'bold', size = 8, color = "black")) +
theme(axis.text.y = element_text(face = 'bold', size = 8, color = 'black')) +
theme(axis.title.x = element_text(face = 'bold', size = 12)) +
theme(axis.title.y = element_text(face = 'bold', size = 12)) +
theme(panel.grid.major = element_blank(), panel.grid.minor = element_blank()) +
theme(legend.text = element_text(face = 'bold', size = 8)) +
theme(legend.title = element_text( face = 'bold', size = 10)) +
theme(strip.text = element_text(face = "bold", size=4))+
theme(legend.position = "none")

```

dU_95



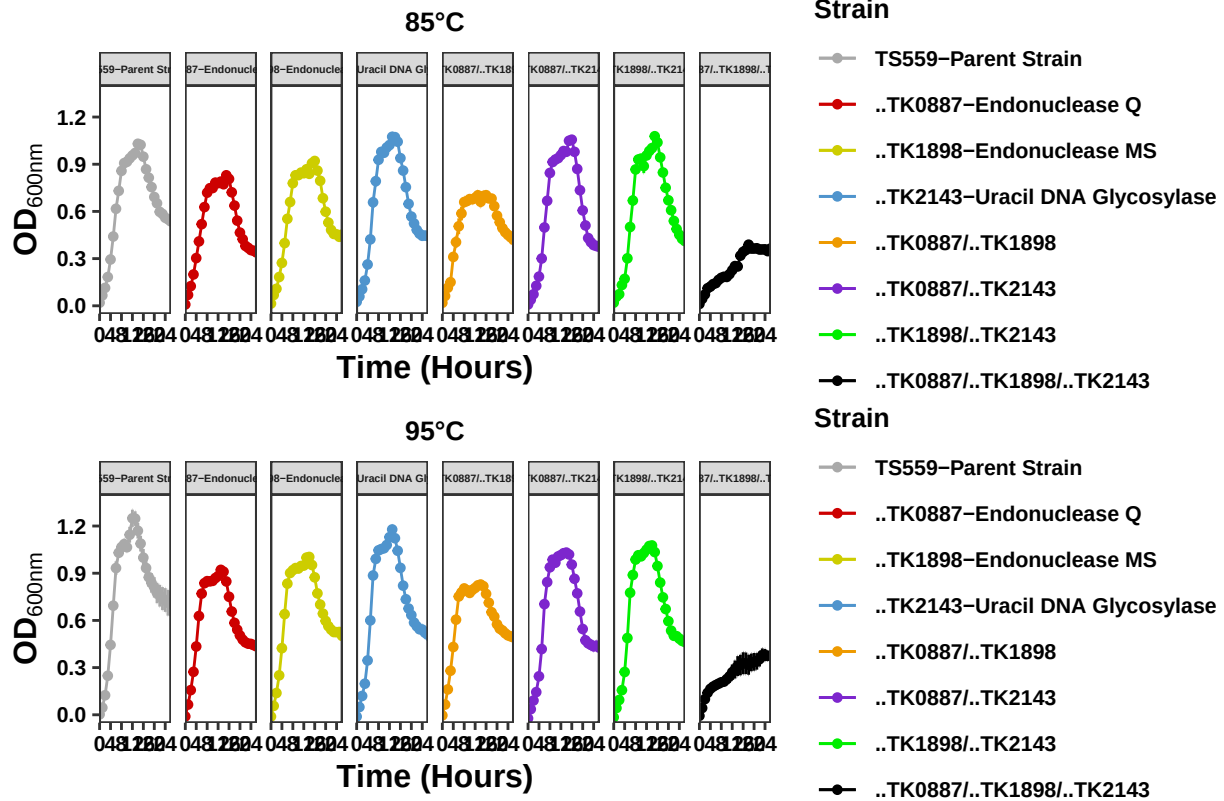
```
#ggarrange
```

```
dU_Plot <- ggarrange(dU_85, dU_95, nrow = 2, legend = "right")
```

```
dU_Plot <- annotate_figure(dU_Plot, top = text_grob("Strains lacking all dU repair components have the most severe fitness defect",
  color = "black", face = "bold", size = 14))
```

```
dU_Plot
```

Strains lacking all dU repair components have the most severe fitness defect



```
ggsave("dU_GC_Final.jpg", dU_Plot, dpi=2000, width = 10, device = 'jpeg')
```

Appendix

```
library(datasets)
library(tidyverse)
library(knitr)
library(ggplot2)
library(ggthemes)
library(ggpubr)
library(ggribes)
```

```

####
#01#
####

dU_GC<-read.csv("Growth_Curve_dU.csv",fill=TRUE,header=TRUE)

head(dU_GC)

####
#02#
####

dU_GC%>%
  ggplot() +
  geom_point(aes(x=Hours, y=Blank.1, color="red")) +
  geom_point(aes(x=Hours, y=Blank.2, color="blue")) +
  geom_point(aes(x=Hours, y=Blank.3, color="green")) +
  geom_point(aes(x=Hours, y=Blank.4, color="purple")) +
  geom_line(aes(x=Hours, y=Blank.1, color="red")) +
  geom_line(aes(x=Hours, y=Blank.2, color="blue")) +
  geom_line(aes(x=Hours, y=Blank.3, color="green")) +
  geom_line(aes(x=Hours, y=Blank.4, color="purple"))

####
#02#
####

GC_85_Blank <-dU_GC %>%
pivot_longer(cols = starts_with("Blank"),
              names_to = "Blank Replicates",
              values_to = "Blank Absorbance",
              values_drop_na = TRUE,
              values_transform = list(Absorbance=as.character)) %>%
select(c(Hours, `Blank Replicates`, `Blank Absorbance`)) %>%
group_by(Hours) %>%
summarize(mean = mean(`Blank Absorbance`), sd= sd(`Blank Absorbance`)) %>%
add_column(Strain = "Blank")

GC_85_TS559 <- dU_GC %>%
  pivot_longer(cols = starts_with("TS559"),
              names_to = "TS559 Replicates",
              values_to = "TS559 Absorbance",
              values_drop_na = TRUE) %>%
select(c(Hours, `TS559 Replicates`, `TS559 Absorbance`)) %>%
group_by(Hours) %>%
summarize(mean = mean(`TS559 Absorbance`), sd = sd(`TS559 Absorbance`)) %>%
add_column(Strain = "TS559")

GC_85_0887 <- dU_GC %>%

```

```

pivot_longer(cols = starts_with("TK0887"),
  names_to = "0887 Replicates",
  values_to = "0887 Absorbance",
  values_drop_na = TRUE) %>%
select(c(Hours, `0887 Replicates`, `0887 Absorbance`)) %>%
group_by(Hours) %>%
summarize(mean = mean(`0887 Absorbance`), sd = sd(`0887 Absorbance`)) %>%
add_column(Strain = "TK0887-Endonuclease Q")

GC_85_1898 <- dU_GC %>%
  pivot_longer(cols = starts_with("TK1898"),
    names_to = "1898 Replicates",
    values_to = "1898 Absorbance",
    values_drop_na = TRUE) %>%
  select(c(Hours, `1898 Replicates`, `1898 Absorbance`)) %>%
  group_by(Hours) %>%
  summarize(mean = mean(`1898 Absorbance`), sd = sd(`1898 Absorbance`)) %>%
  add_column(Strain = "TK1898-Endonuclease MS")

GC_85_2143 <- dU_GC %>%
  pivot_longer(cols = starts_with("TK2143"),
    names_to = "2143 Replicates",
    values_to = "2143 Absorbance",
    values_drop_na = TRUE) %>%
  select(c(Hours, `2143 Replicates`, `2143 Absorbance`)) %>%
  group_by(Hours) %>%
  summarize(mean = mean(`2143 Absorbance`), sd = sd(`2143 Absorbance`)) %>%
  add_column(Strain = "TK2143-Uracil DNA Glycosylase")

GC_85_0887_1898 <- dU_GC %>%
  pivot_longer(cols = starts_with("TK0887.TK1898"),
    names_to = "TK0887/TK1898 Replicates",
    values_to = "TK0887/TK1898 Absorbance",
    values_drop_na = TRUE) %>%
  select(c(Hours, `TK0887/TK1898 Replicates`, `TK0887/TK1898 Absorbance`)) %>%
  group_by(Hours) %>%
  summarize(mean = mean(`TK0887/TK1898 Absorbance`),
    sd = sd(`TK0887/TK1898 Absorbance`)) %>%
  add_column(Strain = "TK0887/TK1898")

GC_85_0887_2143 <- dU_GC %>%
  pivot_longer(cols = starts_with("TK0887.TK2143"),
    names_to = "TK0887/TK2143 Replicates",
    values_to = "TK0887/TK2143 Absorbance",
    values_drop_na = TRUE) %>%
  select(c(Hours, `TK0887/TK2143 Replicates`, `TK0887/TK2143 Absorbance`)) %>%
  group_by(Hours) %>%
  summarize(mean = mean(`TK0887/TK2143 Absorbance`),
    sd = sd(`TK0887/TK2143 Absorbance`)) %>%
  add_column(Strain = "TK0887/TK2143")

GC_85_1898_2143 <- dU_GC %>%

```



```

pivot_longer(cols = starts_with("TK1898.TK2143"),
             names_to = "TK1898/TK2143 Replicates",
             values_to = "TK1898/TK2143 Absorbance",
             values_drop_na = TRUE) %>%
select(c(Hours, `TK1898/TK2143 Replicates`, `TK1898/TK2143 Absorbance`)) %>%
group_by(Hours) %>%
summarize(mean = mean(`TK1898/TK2143 Absorbance`),
          sd = sd(`TK1898/TK2143 Absorbance`)) %>%
add_column(Strain = "TK1898/TK2143")

GC_85_0887_1898_2143<- dU_GC %>%
pivot_longer(cols = starts_with("TK0887.TK1898.TK2143"),
             names_to = "TK0887/TK1898/TK2143 Replicates",
             values_to = "TK0887/TK1898/TK2143 Absorbance",
             values_drop_na = TRUE) %>%
select(c(Hours, `TK0887/TK1898/TK2143 Replicates`,
          `TK0887/TK1898/TK2143 Absorbance`)) %>%
group_by(Hours) %>%
summarize(mean = mean(`TK0887/TK1898/TK2143 Absorbance`),
          sd = sd(`TK0887/TK1898/TK2143 Absorbance`)) %>%
add_column(Strain = "TK0887/TK1898/TK2143")

# combine tables
dU_Table <-rbind(GC_85_Blank, GC_85_TS559)
dU_Table <-rbind(dU_Table, GC_85_0887)
dU_Table <-rbind(dU_Table, GC_85_1898)
dU_Table <-rbind(dU_Table, GC_85_2143)
dU_Table <-rbind(dU_Table, GC_85_0887_1898)
dU_Table <-rbind(dU_Table, GC_85_0887_2143)
dU_Table <-rbind(dU_Table, GC_85_1898_2143)
dU_Table_Final <-rbind(dU_Table, GC_85_0887_1898_2143)

dU_Table_Final %>%
  ggplot(aes(x=Hours, y=mean, color=Strain)) +
  geom_line()
####
#03#
####

dU_GC_tidy<-read.csv("Growth_Curve_dU_tidy.csv",fill=TRUE,header=TRUE)

head(dU_GC_tidy)

dU_GC_tidy_Stat <- dU_GC_tidy %>%
  group_by(Hours, Strain) %>%
  summarize(mean = mean(OD), sd = sd(OD), n=n(), se = sd/sqrt(n)) %>%
  group_by(Strain)

dU_GC_tidy_Stat %>%
  ggplot(aes(x = Hours, y = mean, color=Strain)) +
  geom_line()

```

```

####
#04#
####

dU_GC_tidy_Stat <- dU_GC_tidy_Stat %>%
  mutate(Strain =
    recode(Strain, "TK0887-Endonuclease Q" = "\u0394TK0887-Endonuclease Q",
      "TK1898-Endonuclease MS" = "\u0394TK1898-Endonuclease MS",
      "TK2143-Uracil DNA Glycosylase" = "\u0394TK2143-Uracil DNA Glycosylase",
      "TK0887/TK1898" = "\u0394TK0887/\u0394TK1898",
      "TK0887/TK2143" = "\u0394TK0887/\u0394TK2143",
      "TK1898/TK2143" = "\u0394TK1898/\u0394TK2143",
      "TK0887/TK1898/TK2143" = "\u0394TK0887/\u0394TK1898/\u0394TK2143"))

dU_GC_tidy_Stat%>%
  ggplot(aes(x=Hours, y=mean, color=Strain)) +
  geom_line()

####
#06#
####

dU_GC_tidy_Stat%>%
  ggplot(aes(x=Hours, y= Strain ,color = Strain)) +
  geom_density_ridges_gradient()

dU_GC_tidy_Stat%>%
  ggplot(aes(x=Hours, y=mean, color=Strain, fill=Strain)) +
  geom_step()

dU_GC_tidy_Stat%>%
  ggplot(aes(x=Hours, y=mean, fill=Strain, ymax=mean+sd, ymin=mean-sd)) +
  geom_col(size=1.2, position=position_dodge(preserve='single')) +
  geom_errorbar(width=0.1)
dU_GC_tidy_Stat%>%
  ggplot(aes(x=Hours, y=mean, color=Strain, fill=Strain) ) +
  geom_smooth(se=FALSE)

dU_GC_tidy_Stat%>%
  ggplot(aes(x=Hours, y=mean, color=Strain)) +
  geom_line()

#Perfect!

####
#05b#
####

dU_GC_tidy_Stat%>%

```

```

ggplot(aes(x=Hours, y=mean, color=Strain, ymax=mean+se, ymin=mean-se)) +
  geom_line(size=0.5) +
  geom_point(size=1.2) +
  geom_errorbar(width=0.1)

dU_GC_tidy_Stat%>%
  ggplot(aes(x=Hours, y=mean, color=Strain)) +
  geom_smooth(se=FALSE) +
  geom_point()

####
#06#
####

#help("plotmath")

dU_GC_tidy_Stat%>%
  ggplot(aes(x=Hours, y=mean, color=Strain, ymax=mean+se, ymin=mean-se)) +
  geom_line(size=0.5) +
  geom_point(size=1.2) +
  geom_errorbar(width=0.1) +
  labs(title = "Strains lacking all dU repair components have the most severe fitness defect", subtitle=
    y = "OD600 nm",
    x = "Time (hrs)" ) +
  theme(plot.title = element_text(hjust = 0.5), plot.subtitle= element_text(hjust = 0.5))

dU_GC_tidy_Stat%>%
  ggplot(aes(x=Hours, y=mean, color=Strain, ymax=mean+se, ymin=mean-se)) +
  geom_line(size=0.5) +
  geom_point(size=1.2) +
  geom_errorbar(width=0.1) +
  labs(title = "Strains lacking all dU repair components have the most severe fitness defect", subtitle=
    y = bquote(bold("OD") [600] [nm]),
    x = bquote(bold('Time (hrs)')) ) +
  theme(plot.title = element_text(hjust = 0.5), plot.subtitle= element_text(hjust = 0.5))

####
#07#
####

dU_GC_tidy_Stat%>%
  ggplot(aes(x=Hours, y=mean, color=Strain, ymax=mean+se, ymin=mean-se)) +
  geom_line(size=0.5) +
  geom_point(size=1.2) +
  geom_errorbar(width=0.1) +
  labs(title = "Strains lacking all dU repair components have the most severe fitness defect", subtitle=
    y = bquote(bold("OD") [600] [nm]),
    x = bquote(bold('Time (hrs)')) ) +
  theme(plot.title = element_text(hjust = 0.5), plot.subtitle= element_text(hjust = 0.5)) +
  scale_x_continuous(breaks=c(0, 4, 8, 12, 16, 20, 24), limits=c(0,26)) +

```

```

scale_y_continuous(breaks=c(0, 0.3, 0.6, 0.9, 1.2), limits=c(-0.05, 1.2))

####
#07b#
####

dU_GC_tidy_Stat%>%
  ggplot(aes(x=Hours, y=mean, color=Strain, ymax=mean+se, ymin=mean-se)) +
  geom_line(size=0.5) +
  geom_point(size=1.2) +
  geom_errorbar(width=0.1) +
  labs(title = bquote(bold("Strains lacking all dU repair components have the most severe fitness defect")),
        y = bquote(bold("OD") [600] [nm]),
        x = bquote(bold('Time (hrs)')))) +
  theme(plot.title = element_text(hjust = 0.5), plot.subtitle= element_text(hjust = 0.5)) +
  scale_x_continuous(breaks=c(0, 4, 8, 12, 16, 20, 24), limits=c(0,26), expand = c(0,0)) +
  scale_y_continuous(breaks=c(0, 0.3, 0.6, 0.9, 1.2), limits=c(-0.05, 1.2), expand = c(0,0))

####
#07c#
####

dU_GC_tidy_Stat%>%
  ggplot(aes(x=Hours, y=mean, color=Strain, ymax=mean+se, ymin=mean-se)) +
  geom_line(size=0.5) +
  geom_point(size=1.2) +
  geom_errorbar(width=0.1) +
  labs(title = bquote(bold("Strains lacking all dU repair components have the most severe fitness defect")),
        y = bquote(bold("OD") [600] [nm]),
        x = bquote(bold('Time (hrs)')))) +
  theme(plot.title = element_text(hjust = 0.5), plot.subtitle= element_text(hjust = 0.5)) +
  scale_x_continuous(breaks=c(0, 4, 8, 12, 16, 20, 24), limits=c(0,26), expand = c(0,0)) +
  scale_y_continuous(breaks=c(0, 0.3, 0.6, 0.9, 1.2), limits=c(-0.05, 1.2), expand = c(0,0), trans = "log")

####
#08#
####

dU_GC_tidy_Stat%>%
  ggplot(aes(x=Hours, y=mean, color=Strain, ymax=mean+se, ymin=mean-se)) +
  geom_line(size=0.5) +
  geom_point(size=1.2) +
  geom_errorbar(width=0.1) +
  labs(title = bquote(bold("Strains lacking all dU repair components have the most severe fitness defect")),
        y = bquote(bold("OD") [600] [nm]),
        x = bquote(bold('Time (Hours)')))) +
  theme(plot.title = element_text(hjust = 0.5), plot.subtitle= element_text(hjust = 0.5)) +

```

```

scale_x_continuous(breaks=c(0, 4, 8, 12, 16, 20, 24), limits=c(0,26), expand = c(0,0)) +
scale_y_continuous(breaks=c(0, 0.3, 0.6, 0.9, 1.2), limits=c(-0.05, 1.2), expand = c(0,0))
scale_color_manual(values=c("pink", "brown4", "red3", "yellow3", "steelblue3", "orange2", "purple3",

dU_GC_tidy_Stat%>%
  ggplot() +
  geom_line(aes(x=Hours, y=mean, color=Strain), size=0.5) +
  geom_point(aes(x=Hours, y=mean, color=Strain), size=1.2) +
  geom_errorbar(width=0.1, aes(x=Hours, ymax = mean+sd, ymin = mean-sd, color=Strain)) +
  labs(title = bquote(bold("Strains lacking all dU repair components have the most severe fitness defect
    y = bquote(bold("OD") [600] [nm]),
    x = bquote(bold('Time (hrs)')))) +
    theme(plot.title = element_text(hjust = 0.5), plot.subtitle= element_text(hjust = 0.5)) +
  scale_x_continuous(breaks=c(0, 4, 8, 12, 16, 20, 24), limits=c(0,26), expand = c(0,0)) +
  scale_y_continuous(breaks=c(0, 0.3, 0.6, 0.9, 1.2), limits=c(-0.05, 1.2), expand = c(0,0))
  scale_color_brewer(palette="Set3")

####
#08b#
####

dU_GC_tidy_Stat%>%
  ggplot(aes(x=Hours, y=mean, color=Strain, ymax=mean+se, ymin=mean-se)) +
  geom_line(size=0.5) +
  geom_point(size=1.2) +
  geom_errorbar(width=0.1) +
  labs(title = bquote(bold("Strains lacking all dU repair components have the most severe fitness defect
    y = bquote(bold("OD") [600] [nm]),
    x = bquote(bold('Time (hrs)')))) +
    theme(plot.title = element_text(hjust = 0.5), plot.subtitle= element_text(hjust = 0.5)) +
  scale_x_continuous(breaks=c(0, 4, 8, 12, 16, 20, 24), limits=c(0,26), expand = c(0,0)) +
  scale_y_continuous(breaks=c(0, 0.3, 0.6, 0.9, 1.2), limits=c(-0.05, 1.2), expand = c(0,0)) +
  scale_color_manual(values=c("pink", "brown4", "red3", "yellow3", "steelblue3", "orange2", "purple3",
  geom_errorbar(width=0.1, aes(x=Hours, ymax = mean+sd, ymin = mean-sd, color=Strain))

####
#9#
####

dU_GC_tidy_Stat%>%
  ggplot(aes(x=Hours, y=mean, color=Strain, ymax=mean+se, ymin=mean-se)) +
  geom_line(size=0.5) +
  geom_point(size=1.2) +
  geom_errorbar(width=0.1) +
  labs(title = bquote(bold("Strains lacking all dU repair components have the most severe fitness defect
    y = bquote(bold("OD") [600] [nm]),
    x = bquote(bold('Time (hrs)')))) +
    theme(plot.title = element_text(hjust = 0.5), plot.subtitle= element_text(hjust = 0.5)) +
  scale_x_continuous(breaks=c(0, 4, 8, 12, 16, 20, 24), limits=c(0,26), expand = c(0,0)) +
  scale_y_continuous(breaks=c(0, 0.3, 0.6, 0.9, 1.2), limits=c(-0.05, 1.2), expand = c(0,0)) +
  scale_color_manual(values=c("pink", "brown4", "red3", "yellow3", "steelblue3", "orange2", "purple3",
  geom_errorbar(width=0.1, aes(x=Hours, ymax = mean+sd, ymin = mean-sd, color=Strain)) +
  theme_classic()

```

```
####
#9b#
####
```

```
#using a theme
```

```
dU_GC_tidy_Stat%>%
  ggplot(aes(x=Hours, y=mean, color=Strain, ymax=mean+se, ymin=mean-se)) +
  geom_line(size=0.5) +
  geom_point(size=1.2) +
  geom_errorbar(width=0.1) +
  labs(title = bquote(bold("Strains lacking all dU repair components have the most severe fitness defect")),
        y = bquote(bold("OD") [600] [nm]),
        x = bquote(bold('Time (hrs)')))) +
  theme(plot.title = element_text(hjust = 0.5), plot.subtitle= element_text(hjust = 0.5)) +
  scale_x_continuous(breaks=c(0, 4, 8, 12, 16, 20, 24), limits=c(0,26), expand = c(0,0)) +
  scale_y_continuous(breaks=c(0, 0.3, 0.6, 0.9, 1.2), limits=c(-0.05, 1.2), expand = c(0,0)) +
  scale_color_manual(values=c("pink", "brown", "red3", "yellow3", "steelblue3", "orange2", "purple3", "black")) +
  theme(panel.grid.major = element_blank(), panel.grid.minor = element_blank()) +
  theme_classic()
```

```
#using and adjusting the theme
```

```
dU_GC_tidy_Stat%>%
  ggplot(aes(x=Hours, y=mean, color=Strain, ymax=mean+se, ymin=mean-se)) +
  geom_line(size=0.5) +
  geom_point(size=1.2) +
  geom_errorbar(width=0.1) +
  labs(title = bquote(bold("Strains lacking all dU repair components have the most severe fitness defect")),
        y = bquote(bold("OD") [600] [nm]),
        x = bquote(bold('Time (hrs)')))) +
  scale_x_continuous(breaks=c(0, 4, 8, 12, 16, 20, 24), limits=c(0,26), expand = c(0,0)) +
  scale_y_continuous(breaks=c(0, 0.3, 0.6, 0.9, 1.2), limits=c(-0.05, 1.2), expand = c(0,0)) +
  scale_color_manual(values=c("pink", "brown", "red3", "yellow3", "steelblue3", "orange2", "purple3", "black")) +
  theme_classic() +
  theme(plot.title = element_text(hjust = 0.5, size = 12), plot.subtitle= element_text(hjust = 0.5, size = 10),
        theme(axis.text.x = element_text(face = 'bold', size = 8, color = "black")) +
        theme(axis.text.y = element_text(face = 'bold', size = 8, color = 'black')) +
        theme(axis.title.x = element_text(face = 'bold', size = 12)) +
        theme(axis.title.y = element_text(face = 'bold', size = 12)) +
        theme(panel.grid.major = element_blank(), panel.grid.minor = element_blank()) +
        theme(legend.text = element_text(face = 'bold', size = 8)) +
        theme(legend.title = element_text( face = 'bold', size = 10))
```

```
####
#10#
####
```

```
dU_GC_tidy_Stat%>%
```

```

ggplot(aes(x=Hours, y=mean, color=Strain, ymax=mean+se, ymin=mean-se)) +
  geom_line(size=0.5) +
  geom_point(size=1.2) +
  geom_errorbar(width=0.1) +
  labs(title = bquote(bold("Strains lacking all dU repair components have the most severe fitness defect")),
        y = bquote(bold("OD") [600] [nm]),
        x = bquote(bold('Time (hrs)')))) +
  scale_x_continuous(breaks=c(0, 4, 8, 12, 16, 20, 24), limits=c(0,26), expand = c(0,0)) +
  scale_y_continuous(breaks=c(0, 0.3, 0.6, 0.9, 1.2), limits=c(-0.05, 1.2), expand = c(0,0)) +
  scale_color_manual(values=c("pink", "brown", "red3", "yellow3", "steelblue3", "orange2", "purple3", "black")) +
  theme_classic() +
  theme(plot.title = element_text(hjust = 0.5, size = 12), plot.subtitle= element_text(hjust = 0.5, size = 12),
        axis.text.x = element_text(face = 'bold', size = 8, color = "black")) +
  theme(axis.text.y = element_text(face = 'bold', size = 8, color = 'black')) +
  theme(axis.title.x = element_text(face = 'bold', size = 12)) +
  theme(axis.title.y = element_text(face = 'bold', size = 12)) +
  theme(panel.grid.major = element_blank(), panel.grid.minor = element_blank()) +
  theme(legend.text = element_text(face = 'bold', size = 8)) +
  theme(legend.title = element_text( face = 'bold', size = 10))

ggsave("dU Growth Curve.jpg", dpi=2000, width = 8.25, height = 4, unit = 'in')

####
#11#
####
dU_GC_tidy<-read.csv("Growth_Curve_dU_tidy.csv",fill=TRUE,header=TRUE)

dU_Neg_85 <- dU_GC_tidy %>%
  filter(Strain == "Blank") %>%
  group_by(Hours)%>%
  summarize(blank_mean = mean(OD), .groups = "drop")

dU_GC_tidy_Stat<- dU_GC_tidy %>%
  left_join(dU_Neg_85, by = "Hours") %>%
  mutate(OD = OD - blank_mean) %>%
  group_by(Hours, Strain) %>%
  summarize(mean = mean(OD), sd = sd(OD), n = n(), se = sd / sqrt(n), .groups = "drop") %>%
  filter(Strain != "Blank") %>%
  mutate(Strain =
    recode(Strain, "TK0887-Endonuclease Q" = "\u0394TK0887-Endonuclease Q",
               "TK1898-Endonuclease MS" = "\u0394TK1898-Endonuclease MS",
               "TK2143-Uracil DNA Glycosylase" = "\u0394TK2143-Uracil DNA Glycosylase",
               "TK0887/TK1898" = "\u0394TK0887/\u0394TK1898",
               "TK0887/TK2143" = "\u0394TK0887/\u0394TK2143",
               "TK1898/TK2143" = "\u0394TK1898/\u0394TK2143",
               "TK0887/TK1898/TK2143" = "\u0394TK0887/\u0394TK1898/\u0394TK2143"))

dU_GC_tidy_Stat%>%
  ggplot(aes(x=Hours, y=mean, color=Strain, ymax=mean+se, ymin=mean-se)) +
  geom_line(size=0.5) +
  geom_point(size=1.2) +
  geom_errorbar(width=0.1) +
  theme_bw() +

```

```

labs(title = bquote(bold("Strains lacking all dU repair components have the most severe fitness defect")),
      y = bquote(bold("OD") [600] [nm]),
      x = bquote(bold('Time (Hours)')))) +
scale_x_continuous(breaks=c(0, 4, 8, 12, 16, 20, 24), limits=c(0,26), expand = c(0,0)) +
scale_y_continuous(breaks=c(0, 0.3, 0.6, 0.9, 1.2), limits=c(-0.05, 1.2), expand = c(0,0)) +
scale_color_manual(values=c( "darkgrey", "red3", "yellow3", "steelblue3", "orange2", "purple3", "green3")) +
facet_wrap(~Strain) +
  theme(panel.grid.major = element_blank(), panel.grid.minor = element_blank()) +
  theme(plot.title = element_text(hjust = 0.5, size = 12), plot.subtitle= element_text(hjust = 0.5, size = 12)) +
  theme(axis.text.x = element_text(face = 'bold', size = 8, color = "black")) +
  theme(axis.text.y = element_text(face = 'bold', size = 8, color = 'black')) +
  theme(axis.title.x = element_text(face = 'bold', size = 12)) +
  theme(axis.title.y = element_text(face = 'bold', size = 12)) +
  theme(panel.grid.major = element_blank(), panel.grid.minor = element_blank()) +
  theme(legend.text = element_text(face = 'bold', size = 8)) +
  theme(legend.title = element_text( face = 'bold', size = 10))

ggsave("dU_Plot_85_Facet_Wrapped.jpg", dpi=2000)

#facet_wrap(~factor(Strain, levels=c( "TS559-Parent Strain", "\u0394TK0887-Endonuclease Q", "\u0394TK1887-Endonuclease Q")))

####
#11b#
####

#without the legend

dU_GC_tidy_Stat%>%
  ggplot(aes(x=Hours, y=mean, color=Strain, ymax=mean+se, ymin=mean-se)) +
  geom_line(size=0.5) +
  geom_point(size=1.2) +
  geom_errorbar(width=0.1) +
  theme_bw() +
  labs(title = bquote(bold("Strains lacking all dU repair components have the most severe fitness defect")),
        y = bquote(bold("OD") [600] [nm]),
        x = bquote(bold('Time (Hours)')))) +
scale_x_continuous(breaks=c(0, 4, 8, 12, 16, 20, 24), limits=c(0,26), expand = c(0,0)) +
scale_y_continuous(breaks=c(0, 0.3, 0.6, 0.9, 1.2), limits=c(-0.05, 1.2), expand = c(0,0)) +
scale_color_manual(values=c( "darkgrey", "red3", "yellow3", "steelblue3", "orange2", "purple3", "green3")) +
facet_wrap(~factor(Strain, levels=c( "TS559-Parent Strain", "\u0394TK0887-Endonuclease Q", "\u0394TK1887-Endonuclease Q"))) +
  theme(panel.grid.major = element_blank(), panel.grid.minor = element_blank()) +
  theme(plot.title = element_text(hjust = 0.5, size = 12), plot.subtitle= element_text(hjust = 0.5, size = 12)) +
  theme(axis.text.x = element_text(face = 'bold', size = 8, color = "black")) +
  theme(axis.text.y = element_text(face = 'bold', size = 8, color = 'black')) +
  theme(axis.title.x = element_text(face = 'bold', size = 12)) +
  theme(axis.title.y = element_text(face = 'bold', size = 12)) +
  theme(panel.grid.major = element_blank(), panel.grid.minor = element_blank()) +
  theme(legend.text = element_text(face = 'bold', size = 8)) +
  theme(legend.title = element_text( face = 'bold', size = 10)) +
  theme(strip.text = element_text(face = "bold", size=6))+
  theme(legend.position = "none")

```



```

ggsave("dU Growth Curve_Facet_Wrapped.jpg", dpi = 2000)

#without the panel labels

dU_GC_tidy_Stat%>%
  ggplot(aes(x=Hours, y=mean, color=Strain, ymax=mean+se, ymin=mean-se)) +
  geom_line(size=0.5) +
  geom_point(size=1.2) +
  geom_errorbar(width=0.1) +
  theme_bw() +
  labs(title = bquote(bold("Strains lacking all dU repair components have the most severe fitness defects")),
       y = bquote(bold("OD") [600] [nm]),
       x = bquote(bold('Time (Hours)')))) +
  scale_x_continuous(breaks=c(0, 4, 8, 12, 16, 20, 24), limits=c(0,26), expand = c(0,0)) +
  scale_y_continuous(breaks=c(0, 0.3, 0.6, 0.9, 1.2), limits=c(-0.05, 1.2), expand = c(0,0)) +
  scale_color_manual(values=c( "darkgrey", "red3", "yellow3", "steelblue3", "orange2", "purple3", "green3")) +
  facet_wrap(~factor(Strain, levels=c( "TS559-Parent Strain", "\u0394TK0887-Endonuclease Q", "\u0394TK1111-Endonuclease Q"))) +
  theme(panel.grid.major = element_blank(), panel.grid.minor = element_blank()) +
  theme(plot.title = element_text(hjust = 0.5, size = 12), plot.subtitle= element_text(hjust = 0.5, size = 10)) +
  theme(axis.text.x = element_text(face = 'bold', size = 8, color = "black")) +
  theme(axis.text.y = element_text(face = 'bold', size = 8, color = 'black')) +
  theme(axis.title.x = element_text(face = 'bold', size = 12)) +
  theme(axis.title.y = element_text(face = 'bold', size = 12)) +
  theme(panel.grid.major = element_blank(), panel.grid.minor = element_blank()) +
  theme(legend.text = element_text(face = 'bold', size = 8)) +
  theme(legend.title = element_text( face = 'bold', size = 10)) +
  theme( strip.text.x = element_blank() )

ggsave("dU Growth Curve_Facet_Wrapped_2.jpg", dpi=2000, width = 12)

####
#12#
####

#make 85 degree data into an item

dU_85 <- dU_GC_tidy_Stat%>%
  ggplot(aes(x=Hours, y=mean, color=Strain, ymax=mean+se, ymin=mean-se)) +
  geom_line(size=0.5) +
  geom_point(size=1.2) +
  geom_errorbar(width=0.1) +
  theme_bw() +
  labs(subtitle=bquote(bold("85\u00B0C")),
       y = bquote(bold("OD") [600] [nm]),
       x = bquote(bold('Time (Hours)')))) +
  scale_x_continuous(breaks=c(0, 4, 8, 12, 16, 20, 24), limits=c(0,26), expand = c(0,0)) +
  scale_y_continuous(breaks=c(0, 0.3, 0.6, 0.9, 1.2), limits=c(-0.05, 1.4), expand = c(0,0)) +
  scale_color_manual(values=c( "darkgrey", "red3", "yellow3", "steelblue3", "orange2", "purple3", "green3")) +
  facet_wrap(ncol=8, ~factor(Strain, levels=c( "TS559-Parent Strain", "\u0394TK0887-Endonuclease Q", "\u0394TK1111-Endonuclease Q"))) +
  theme(panel.grid.major = element_blank(), panel.grid.minor = element_blank()) +
  theme(plot.title = element_text(hjust = 0.5, size = 12), plot.subtitle= element_text(hjust = 0.5, size = 10))

```

```

theme(axis.text.x = element_text(face = 'bold', size = 8, color = "black")) +
theme(axis.text.y = element_text(face = 'bold', size = 8, color = 'black')) +
theme(axis.title.x = element_text(face = 'bold', size = 12)) +
theme(axis.title.y = element_text(face = 'bold', size = 12)) +
theme(panel.grid.major = element_blank(), panel.grid.minor = element_blank()) +
theme(legend.text = element_text(face = 'bold', size = 8)) +
theme(legend.title = element_text( face = 'bold', size = 10)) +
theme(strip.text = element_text(face = "bold", size=4))+
theme(legend.position = "none")

```

dU_85

#95 degree data

```
dU_GC_tidy_95<-read.csv("Growth_Curve_dU_tidy_95.csv",fill=TRUE,header=TRUE)
```

```

dU_Neg_95 <- dU_GC_tidy_95 %>%
  filter(Strain == "Blank") %>%
  group_by(Hours)%>%
  summarize(blank_mean = mean(OD), .groups = "drop")

dU_GC_tidy_Stat_95<- dU_GC_tidy_95 %>%
  left_join(dU_Neg_95, by = "Hours") %>%
  mutate(OD = OD - blank_mean) %>%
  group_by(Hours, Strain) %>%
  summarize(mean = mean(OD), sd = sd(OD), n = n(), se = sd / sqrt(n), .groups = "drop") %>%
  filter(Strain != "Blank") %>%
  mutate(Strain =
    recode(Strain, "TK0887-Endonuclease Q" = "\u0394TK0887-Endonuclease Q",
      "TK1898-Endonuclease MS" = "\u0394TK1898-Endonuclease MS",
      "TK2143-Uracil DNA Glycosylase" = "\u0394TK2143-Uracil DNA Glycosylase",
      "TK0887/TK1898" = "\u0394TK0887/\u0394TK1898",
      "TK0887/TK2143" = "\u0394TK0887/\u0394TK2143",
      "TK1898/TK2143" = "\u0394TK1898/\u0394TK2143",
      "TK0887/TK1898/TK2143" = "\u0394TK0887/\u0394TK1898/\u0394TK2143"))

```

```

dU_95 <- dU_GC_tidy_Stat_95%>%
  ggplot(aes(x=Hours, y=mean, color=Strain, ymax=mean+se, ymin=mean-se)) +
  geom_line(size=0.5) +
  geom_point(size=1.2) +
  geom_errorbar(width=0.1) +
  theme_bw() +
  labs( subtitle=bquote(bold("95\u00B0C")),
    y = bquote(bold("OD") [600] [nm]),
    x = bquote(bold('Time (Hours)')))) +
  scale_x_continuous(breaks=c(0, 4, 8, 12, 16, 20, 24), limits=c(0,26), expand = c(0,0)) +
  scale_y_continuous(breaks=c(0, 0.3, 0.6, 0.9, 1.2), limits=c(-0.05, 1.4), expand = c(0,0)) +
  scale_color_manual(values=c( "darkgrey", "red3", "yellow3", "steelblue3", "orange2", "purple3", "green3")) +
  facet_wrap(ncol=8, ~factor(Strain, levels=c( "TS559-Parent Strain", "\u0394TK0887-Endonuclease Q", "\u0394TK1898-Endonuclease MS", "\u0394TK2143-Uracil DNA Glycosylase", "TK0887/TK1898", "TK0887/TK2143", "TK1898/TK2143", "TK0887/TK1898/TK2143"))) +
  theme(panel.grid.major = element_blank(), panel.grid.minor = element_blank()) +
  theme(plot.title = element_text(hjust = 0.5, size = 12), plot.subtitle= element_text(hjust = 0.5, size = 10)) +
  theme(axis.text.x = element_text(face = 'bold', size = 8, color = "black")) +

```

```

theme(axis.text.y = element_text(face = 'bold', size = 8, color = 'black')) +
theme(axis.title.x = element_text(face = 'bold', size = 12)) +
theme(axis.title.y = element_text(face = 'bold', size = 12)) +
theme(panel.grid.major = element_blank(), panel.grid.minor = element_blank()) +
theme(legend.text = element_text(face = 'bold', size = 8)) +
theme(legend.title = element_text( face = 'bold', size = 10)) +
theme(strip.text = element_text(face = "bold", size=4))+
theme(legend.position = "none")
dU_95
#ggarrange

dU_Plot <- ggarrange(dU_85, dU_95, nrow = 2, legend = "right")

dU_Plot <-annotate_figure(dU_Plot, top = text_grob("Strains lacking all dU repair components have the m
          color = "black", face = "bold", size = 14))

dU_Plot

ggsave("dU_GC_Final.jpg", dU_Plot, dpi=2000, width = 10, device = 'jpeg')

```